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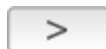
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VSMDefTopic



About the VSM

About the VSM

The Voltage Source Module (VSM) instrument is a dual-output high-current power supply that allows you to test VLSI devices with high power requirements. The VSM has two 81A DCVS channels that can be used separately or in parallel to get a 324W power source that can provide up to 162A of current. In addition, separate modules can be merged together to provide even more current. Specific restrictions on merging apply, but up to three modules (six supplies) can be merged together for 486A of current as a single supply. For ease of use, the VSM contains both an internal calibration and diagnostic functionality using embedded control code.

This section includes the following topics:

- [Features](#)
- [Safety Guidelines](#)
- [Key Terms](#)
- [VSM Calibration](#)
- [VSM Software Licensing Requirements](#)

The VSM document includes the following sections:

- [VSM Theory of Operation](#)
- [VSM DCVS Programming](#)
- [DCVS and VSM Board Debug Displays](#)
- [VSM Examples](#)

Related Topics

- [VSM Specifications](#)

- [DCVS](#) (VBT syntax)
-
- [VSM DIB Design Guidelines](#)

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About the VSM : Features

Features

For VSM hardware specifications, see [VSM Specifications](#).

VSM Hardware

The VSM hardware includes the following features:

- | | |
|--|---|
| | <ul style="list-style-type: none">Two 81A power supplies that can be used in parallel for a combined 162A of current to the DUT |
| | <ul style="list-style-type: none">Modules that can be merged to provide up to 486A of current to the DUT (three modules with six supplies) |
| | <ul style="list-style-type: none">Single output voltage range of 0V to 4V |
| | <ul style="list-style-type: none">Multiple output current ranges, as shown in table below |
| | <ul style="list-style-type: none">Multiple current measurement ranges, as shown in table below |
| | <ul style="list-style-type: none">Maximum DC output current (81A) limited at output voltages over 2V to a maximum power of 162W per supply, or 324W per VSM |
| | <ul style="list-style-type: none">Less than 50mV of droop with 1000F of load board capacitance (81 A supply) |

Adjustable supply compensations (bandwidths) support load capacitances of from 40F @ 1A or 11A mode to greater than 2,000F maximum per 81A supply. Droop specification is a direct function of

DIB capacitance and bandwidth settings.

- A special 1.0A output current range for fixture protection
- Supports up to 128 PSets: Force-V, Clamp-I, I-Modes, and measure RAM sample frequency. (PSets can be controlled from the pattern generator or applied using VBT language.)
- Vbump capability to allow pattern generator selection of any of two force voltage values
- A safety interlock that does not allow output power without the presence of a VSM-designed DIB
- Built-in calibration capability, with NIST-traceable calibration elements. No external fixtures are required for calibration
- Voltage outputs on the DIB proportional to voltage (Vout) and current (Iout) to be used on the DIB for DUT temperature control
- Ability to sink limited DC current (minimum of 5% of current mode range) indefinitely and support full source and sink current dynamically
- Over 100 capture signal setups available with 32 usable in any one burst
- Allows measurements triggered by pattern generator control and stored in measure RAM (256K), providing current or voltage profiling capability
- A complete set of timed current limiting features, with two different programmable times and levels
- Built-in validation capability, with static and dynamic loads available to verify specification performance at the module level
- Alarms for Kelvin drops between the power supply and DIB board, and user programmable Kelvin alarm drops between the DIB board and DUT.

VSM Output Current and Current Measurement Ranges

VSM DC Output Current and Current Measurement Ranges

Configuration ¹	Output Current Range (Amps)	Current Measurement Ranges	
		125% (Amps)	250% (Amps)
One supply: One VSM board, one channel	1	1.25	2.5
	11	13.75	27.5
	21	26.25	52.5
	51	63.75	127.5
	81	101.25	202.5
Two supplies: One VSM board, merging two channels	1	1.25	2.5
	2	2.5	5
	11	13.75	27.5
	21	26.25	52.5
	51	63.75	127.5
	81	101.25	202.5
	102	127.5	255
	162	202.5	405
Three supplies: Two VSM boards, merging two channels on first board and one on second board	1	1.25	2.5
	3	3.75	7.5
	11	13.75	27.5
	21	26.25	52.5
	51	63.75	127.5
	81	101.25	202.5
	153	191.25	382.5
	243	303.75	607.5
Four supplies: Two VSM boards, merging two channels on	1	1.25	2.5
	4	5	10
	11	13.75	27.5

first board and two on second board	21	26.25	52.5
	51	63.75	127.5
	81	101.25	202.5
	204	255	510
	324	405	810
Five supplies:	1	1.25	2.5
Three VSM boards, merging two channels on first board, two on second board, and one on third board	5	6.25	12.5
	11	13.75	27.5
	21	26.25	52.5
	51	63.75	127.5
	81	101.25	202.5
	105	131.25	262.5
	255	318.75	637.5
	405	506.25	1012.5
Six supplies:	1	1.25	2.5
Three VSM boards, merging two channels on first board, two on second board, and two on third board	6	7.5	15
	11	13.75	27.5
	21	26.25	52.5
	51	63.75	127.5
	81	101.25	202.5
	126	157.5	315
	306	382.5	765
	486	607.5	1215

¹ When merging VSM channels, the first two merged channels must be on the same board. The third, fourth, fifth, and sixth channels can be added from any other boards in any combination. That is, four channels do not have to come from two boards.

DCVS Instruments

Each of the two VSM DCVS instruments can accurately force a programmed voltage level and measure power supply current to the DUT and has the following features.

DCVS Instrument Feature Highlights

Feature	Specification
DCVS channels	Two per board
Merging capability	Up to six channels
Maximum DC current output (single channel)	81A (162W)
Maximum DC current output (merged channels)	2 VSM channels: 162A (324W) 3 VSM channels: 243A (486W) 4 VSM channels: 324A (648W) 5 VSM channels: 405A (810W) 6 VSM channels: 486A (972W)
Force voltage range	0 to 4.0V
Current measurement ranges	See VSM Output Current and Current Measurement Ranges
Voltage measurement ranges	3.0V and 6.0V
Measure current filters	400kHz, 40.625kHz, 2.529kHz, 635Hz
Current range source	0 to 81A per channel
Current range sink	0 to a minimum of -4A per channel
Pattern control	Output levels, current measurement, PSets, captures, condition bit control, and more. See DCVS Pattern Control
Current limiting	Programmable multilevel overload and fold current limits (5-110% of the output current range)
Compensation bandwidth	Programmable transient compensation bandwidth (five settings)
Capture memory depth	256K of 12-bit current or voltage measurements
Capture memory sampling rates	15.2587kHz to 31.25MHz. Can be set to a value equal to $31.25M/2^n$, where $n = 0$ through 12.

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About the VSM : Safety Guidelines

Safety Guidelines

As defined by IEC standards, an energy hazard exists when power levels exceed 240W. This type of hazard can produce sparks and cause conductive parts to weld to electrically conductive surfaces if they come in contact with the high-current sources. By this definition, the VSM board installed on the UltraFLEX test system represents an electrical hazard both to operators and equipment. Each of the two voltage sources on the VSM can supply a maximum of 162W (a value that can be obtained through various current and voltage combinations such as, 81A at 2.0V). Merging these voltage sources together supplies up to 324W (for example, 162A at 2.0V). The VSM is powered by +48V with a maximum of 4V distributed to the test head terminals. For complete safety information, see [About Operator and Safety Information](#).

Warning:

Because of the high current available from the VSM instrument at the test head terminals, follow the safety precautions listed below to avoid injuries due to electric shock or burns and to prevent damage to the equipment such as melting or burning of components and boards.

Observe the following safety precautions when using the VSM.

- | | |
|---|--|
| • | Advise operators of the potential energy hazard and ask them to exercise caution during testing, especially if two or more high-current voltage sources are operating in parallel. |
| • | Do not touch the test head interface during testing or when the interface is powered on. |
| • | Avoid using conductive tools when using the test system and do not wear rings, watches, bracelets, necklaces, or other objects that are metallic or conductive. |

- Exercise caution when using conductive tools or probes. Make sure that these tools or probes do not come in contact with the high current sources or any other conductive part of the system.
- Exercise caution when probing on the test head to avoid inadvertent shorts between voltage or current terminals. Watch out especially for situations where a probe could become loose and touch two voltage or current terminals.
- Wear antistatic wrist straps with internal current limiting resistor because the VSM is subject to static discharge.
- Wear safety glasses.
- Exercise caution when designing and running tests and when setting current limits for tests. You may see large current changes by simply changing voltage levels or turning the large current-consuming elements of a device on and off. Verify the ripple current specifications for the DIB capacitors to meet these rapid current changing conditions. This is true any time the voltage is changing rapidly during Vbump, shmoo, and other testing activities.
- Keep the high-current areas clear of any items that might alter their contacts. They must be clean and free of liquid or other conductors. Resistance from poor electrical contact may also result from physical damage, wear, or operation.
- Inspect the high-current area frequently and repair unsafe DIBs before use.
- Affix a "DANGER ELECTRICAL HAZARD" warning label to any DIB that uses the VSM instrument, including customer-designed DIBs.
- Make sure that DIBs using the VSM option have a protective cover with recessed or protected probe points, if present, to prevent operators from inadvertently touching the high-current components or dropping conductive tools or equipment onto them.
- Have an emergency response plan that all operators are familiar with. If an emergency occurs, immediately stop testing, get medical help if needed, and report the emergency to proper site

personnel. Do not touch the test head as items may be burning, hot, or still electrically charged.

- If you detect smoldering, sparks, component burning or melting, or other severe failure, power down the test system immediately by pressing an EMO button. Do not touch any of the damaged parts as they may still be very hot.

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About the VSM : Key Terms

Key Terms

This section uses the following terms, abbreviations and acronyms:

capture	The process of storing many individual voltage or current measurements taken in real time at a constant sample rate over a period of time.
capture memory	The VSM hardware used to store sampled values. The VSM can capture up to 256K samples at a sample rate of up to 31.25M samples/second.
current profile	The plot of sampled current measurements over time.
DCVS	Direct current voltage supply. This term is used to refer to each channel of the VSM board. Each DCVS on the VSM board is an instrument with independently programmable voltage output, current limits, voltage or current meter, capture memory, and parameter set memories.
DGS	Device ground sense.
force line	Wiring and DIB interposer that carry the voltage supply current to the DIB power planes and DUT.
HSSL	High-speed serial link, a 625Mbps serial databus used for command register communications.
ISL	Instrument Sync-Link, a high-speed (2.5Gbps) serial bus that carries PatGen data between instruments and provides for board-to-board communications through the backplane. The PatGen data has a timestamp embedded in the pattern that tells the instrument when to execute a given command or pattern burst to the DUT.

PatGen	The VSM pattern generator can be used to synchronize VSM capabilities like capture, strobe, and Vbump to other instruments in the same PatGen domain. The PatGen controls the “real-time cadence of events” in a sequence.
PSet	A parameter set that sets the instrument state during a pattern burst.
sense line	An electrical connection used to measure the voltage at the Kelvin point. This connection is used to control the output of the force line to maintain the programmed output voltage at the desired level.
Vbump test	A test that changes the device power pin (VDD) level during operation to see if you can read from and write to the device correctly at different voltage levels. Vbump is also used to describe whenever the programmed voltage level for a device power supply is changed during a test pattern.
VSG	Voltage supply ground. This is the overall ground plane within the VSM. This is equivalent to the digital ground plane on other UltraFLEX boards.
VSM	Voltage Source Module. The VSM has two 81A channels that can be used separately or in parallel to get a 162A power source. In addition, separate modules can be merged together to provide even more current.

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About the [VSM](#) : VSM Calibration

VSM Calibration

VSM calibration is handled internally by IG-XL and does not require the use of a calibration DIB.

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About the [VSM](#) : VSM Software Licensing Requirements

VSM Software Licensing Requirements

Operation of the VSM does not require a software license.

For more information on IG-XL licensing, see [IG-XL Licensing](#) in IG-XL Administration help.

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VSM Examples

VSM Examples

This section contains two kinds of program examples for the VSM:

- [IG-XL workbook examples](#) show all IG-XL data, VBT code, and patterns (if used) necessary for a complete running program.
- [VBT function examples](#) show one or more VBT functions that perform a programmed action when called as specified.

Workbook Examples

- [VSM Static Icc Programming](#)

VBT Function Examples

- [Basic DCVS Setup Programming](#)
- [Vbump Testing](#)
- [Current Meter Setup, Strobe, and Readback Programming](#)
- [Current Profiling](#)
- [DCVS Current Range and Limit Programming](#)

Related Topics

- [DCVS \(VBT Syntax Reference\)](#)

- [VSM DCVS Programming](#)

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VSM Examples : VSM Static Icc Programming

VSM Static Icc Programming

This workbook example shows how to program a VSM static Icc test that forces voltage and measures current (FVMI) on two SN74ALS245AN bidirectional digital transceiver devices on two sites. This example connects one VSM channel to the Vcc pin, applies 4.0V to the pin, and measures the resulting current while the digital pins are set to a known state. This example uses one HSD1000 board in slot 16 and a VSM board in slot 19.

This workbook example includes the following program components:

- | | |
|---|-----------------|
| • | DUT Connections |
| • | DataTool Sheets |
| • | Patterns |
| • | VBt Code |

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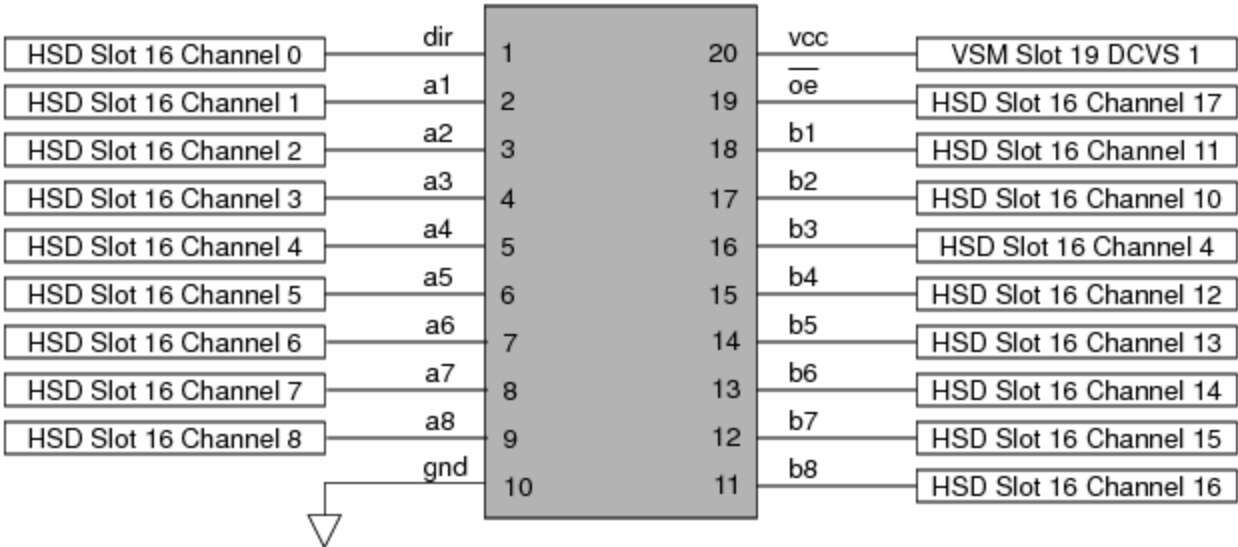
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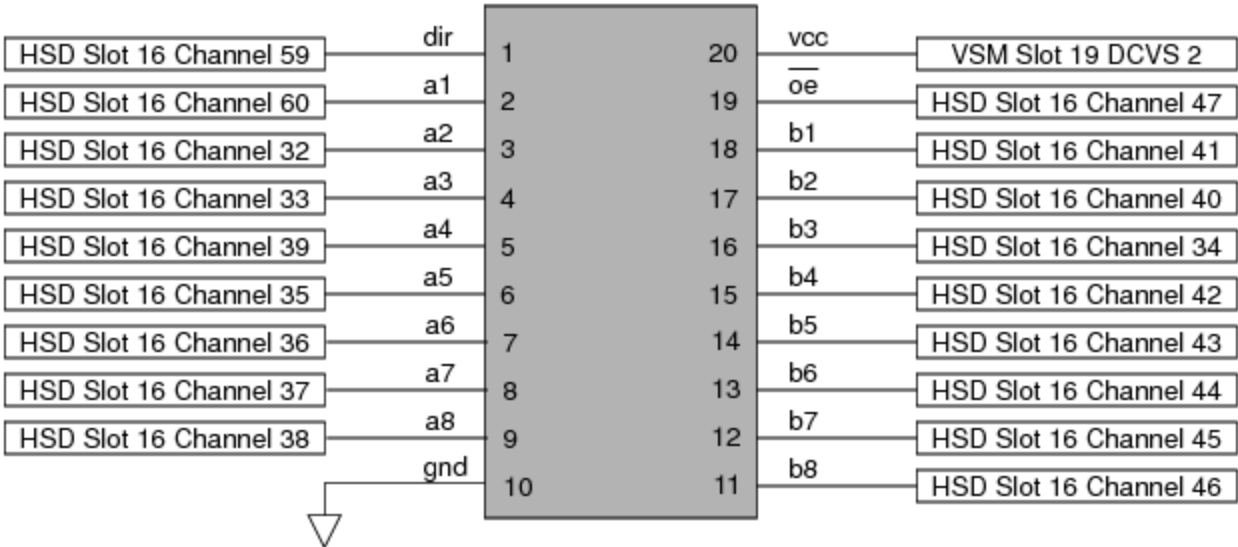
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VSM Examples : VSM Static Icc Programming : DUT Connections

DUT Connections



HSD1000 Connections for Digital Transceiver Device Site 0



HSD1000 Connections for Digital Transceiver Device Site 1

		
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VSM Examples : VSM Static Icc Programming : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- | | |
|---|----------------------|
| • | Pin Map Sheet |
| • | Channel Map Sheet |
| • | Pin Levels Sheet |
| • | Global Specs Sheet |
| • | Test Instances Sheet |
| • | Flow Table Sheet |

Pin Map Sheet

Defining the pins:

Pin Map

Group Name	Pin Name	Type
	dir	Input
	a1	I/O
	a2	I/O
	a3	I/O
	a4	I/O
	a5	I/O
	a6	I/O
	a7	I/O
	a8	I/O
	gnd	Gnd
	b8	I/O
	b7	I/O
	b6	I/O
	b5	I/O
	b4	I/O
	b3	I/O
	b2	I/O
	b1	I/O
	oe	Input
	vcc	Power

Defining the pin groups:

Pin Map

Group Name	Pin Name	Type
ctrls	oe	Input
ctrls	dir	
porta	a1	I/O
porta	a2	
porta	a3	
porta	a4	
porta	a5	
porta	a6	
porta	a7	
porta	a8	
portb	b1	I/O
portb	b2	
portb	b3	
portb	b4	
portb	b5	
portb	b6	
portb	b7	
portb	b8	
ports	porta	I/O
ports	portb	
digital_pins	ctrls	I/O
digital_pins	ports	

[Channel Map Sheet](#)

Channel Map

DIB ID:

View Mode:

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Site 1
a5	Pin 6	I/O	16.ch5	16.ch35
a6	Pin 7	I/O	16.ch6	16.ch36
a7	Pin 8	I/O	16.ch7	16.ch37
a8	Pin 9	I/O	16.ch8	16.ch38
gnd	Pin 10	N/C		
b1	Pin 18	I/O	16.ch11	16.ch41
b2	Pin 17	I/O	16.ch10	16.ch40
b3	Pin 16	I/O	16.ch4	16.ch34
b4	Pin 15	I/O	16.ch12	16.ch42
b5	Pin 14	I/O	16.ch13	16.ch43
b6	Pin 13	I/O	16.ch14	16.ch44
b7	Pin 12	I/O	16.ch15	16.ch45
b8	Pin 11	I/O	16.ch16	16.ch46
oe	Pin 19	I/O	16.ch17	16.ch47
vcc	Pin 20	DCVS	19.sense1	19.sense2

Pin Levels Sheet

Defined using formulas:

Pin Levels

Pin/Group	Seq.	Parameter	Value
vcc	0	Vps	4
vcc		Isc	=_gl_lsc
vcc		Tdelay	0.1
ctrls		Vil	0
ctrls		Vih	5
ctrls		Vt	2.5
ctrls		Vcl	=_Vcl_default
ctrls		Vch	=_Vch_default
ctrls		DriverMode	Largeswing-HiZ
ports		Vil	0
ports		Vih	5
ports		Vol	0.8
ports		Voh	2.4
ports		Vt	=0
ports		Vcl	=_Vcl_default
ports		Vch	=_Vch_default
ports		VoutLoTyp	0.1
ports		VoutHiTyp	4.9
ports		DriverMode	Largeswing-VT1K

Formulas evaluated:

Pin Levels

Pin/Group	Seq.	Parameter	Value
vcc	0	Vps	4
vcc		Isc	1
vcc		Tdelay	0.1
ctrls		Vil	0
ctrls		Vih	5
ctrls		Vt	2.5
ctrls		Vcl	-1
ctrls		Vch	6
ctrls		DriverMode	Largeswing-HiZ
ports		Vil	0
ports		Vih	5
ports		Vol	0.8
ports		Voh	2.4
ports		Vt	0
ports		Vcl	-1
ports		Vch	6
ports		VoutLoTyp	0.1
ports		VoutHiTyp	4.9
ports		DriverMode	Largeswing-VT1K

Global Specs Sheet

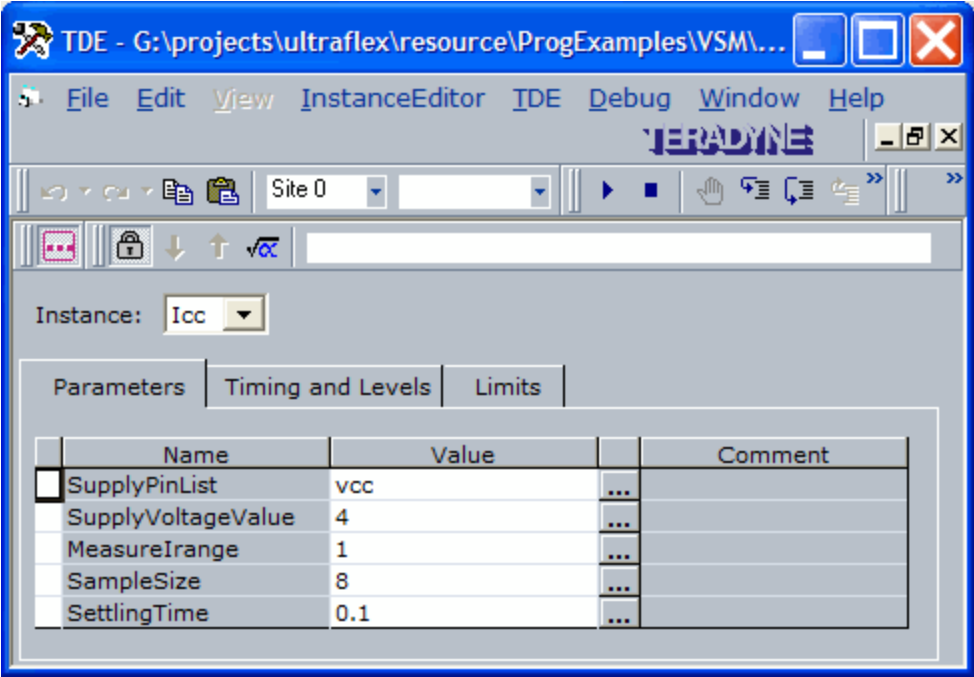
Global Specs

Symbol	Job	Value	Comment
Vcl_default		-1	Detector clamp voltage low
Vch_default		6	Detector clamp voltage high
gl_Isc		1	
gl_Tdelay		=20*ms	

Test Instances Sheet

Test Instances		TERADYNE									
Test Procedure				DC Specs		AC Specs		Sheet Parameters			
Test Name	Type	Name	Called As	Category	Selector	Category	Selector	Time Sets	Edge Sets	Pin Levels	Mixed Signal Timing
Icc	VBT	Icc_VBT								Levels	

The Icc test instance has the following test-specific parameters and values as shown in the Instance Editor:



Flow Table Sheet

Flow Table

		Flow Domain:																	
		Gate		Command				Limits		Datalog Display		Bin Number		Sort Number					
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim	Scale	Units	Format	Pass	Fail	Pass	Fail	Result	
					set-error-bin										99		999		
					Test	lcc													
					Use-Limit	lcc			0.025	0.085					4		40	Fail	
					set-device									1		1		Pass	

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VSM Examples : [VSM Static lcc Programming](#) : Patterns

Patterns

This program example does not require any pattern files.

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VSM Examples : [VSM Static Icc Programming](#) : VBT Code

VBT Code

Option Explicit

```
Public Function Icc_VBT(SupplyPinList As PinList, _
    SupplyVoltageValue As Double, _
    MeasureIrange As Double, _
    SampleSize As Long, _
    SettlingTime As Double) As Long
```

```
Dim Measure As New PinListData
```

```
On Error GoTo errHandler
```

```
' Apply HSD levels and power supply pin values.
```

```
TheHdw.Digital.ApplyLevelsTiming True, True, False, tlPowered
```

```
With TheHdw.DCVS.Pins(SupplyPinList)
```

```
    ' Override actual programmed SupplyVoltageValue if required.
```

```
    If SupplyVoltageValue <> 0 Then
```

```
        ' Use specified SupplyVoltageValue for test.
```

```
        .Voltage.Main.Value = SupplyVoltageValue
```

```
    Else
```

```
        ' Read back actual programmed force level value for datalog.
```

```
        SupplyVoltageValue = .Voltage.Main.Value
```

```
    End If
```

```
' - - - Set up the DCVS instrument to perform lcc measurement. - - -  
' The current limit is set within the ApplyLevelsTiming statement.  
' (LoadLevels = True)  
.Meter.Mode = tlDCVSMeterCurrent  
.Meter.CurrentRange.Value = MeasureIrange
```

TheHdw.Wait (SettlingTime)

```
' Strobe meter and read back samples.  
Measure = .Meter.Read(tlStrobe, SampleSize)
```

End With

```
' - - - Datalog Results - - -  
TheExec.Flow.TestLimit Measure, , , tlSignGreaterEqual, _  
tlSignLessEqual, scaleNone, unitAmp, , _  
TheExec.DataManager.instanceName, _  
CompareAverage, SupplyPinList, _  
SupplyVoltageValue, unitVolt, , , tlForceFlow
```

Exit Function

```
errHandler:  
If AbortTest Then Exit Function Else Resume Next
```

End Function

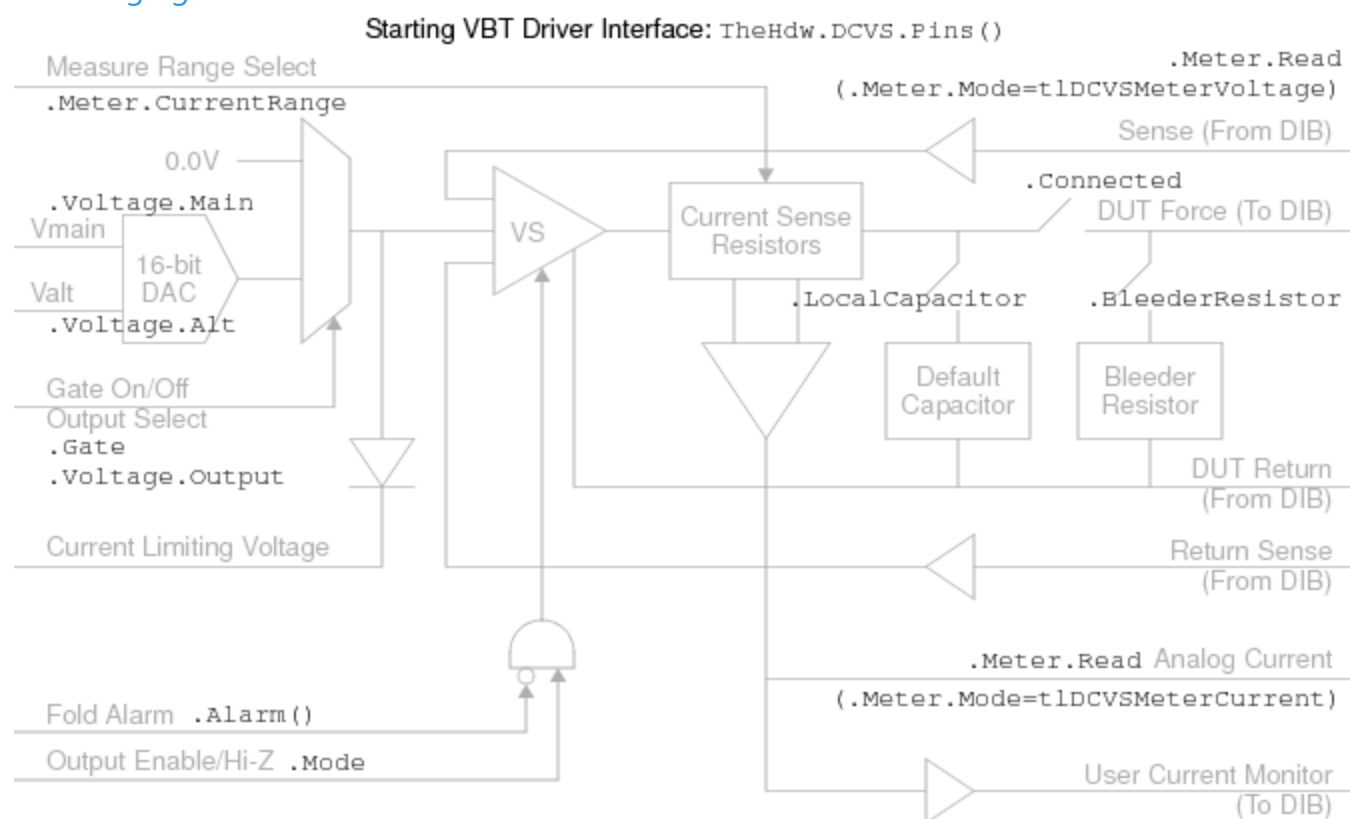
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VSM Examples : Basic DCVS Setup Programming

Basic DCVS Setup Programming

The following example describes how to program the main output subsystem as shown in the following figure.



DCVS Instrument: Programming the Main Output Subsystem

Sub DCVS_BasicSetup()

With TheHdw.DCVS.Pins("VDD")

' Set the DCVS bandwidth to optimize for desired transient response.

```
.BandwidthSetting = 3
```

' Auto is the default setting for the bleeder resistor and local capacitor,
' but it may be turned on or off for some applications.

.BleederResistor = tIDCVSAuto

.LocalCapacitor = tIDCVSAuto

' You can force merged DCVS into a lower current output mode. The lower
' current output mode is necessary for tighter current limiting
' specifications and low current metering.

.CurrentRange.Value = .CurrentRange.Min

' Set the main and alt voltages

.Voltage.Main.Value = 2.5

.Voltage.Alt.Value = 2.1

' Set the output to Main DAC value. You can select Alt later using VBT or a
' pattern opcode

.Voltage.Output = tIDCVSVoltageMain

' Set the DCVS output mode to Voltage (not High Impedance)

.Mode = tIDCVSModeVoltage

' Connect the DCVS output (This connects both force and sense)

.Connect (tIDCVSConnectDefault)

' Gate on the DCVS

.Gate = True

End With

End Sub



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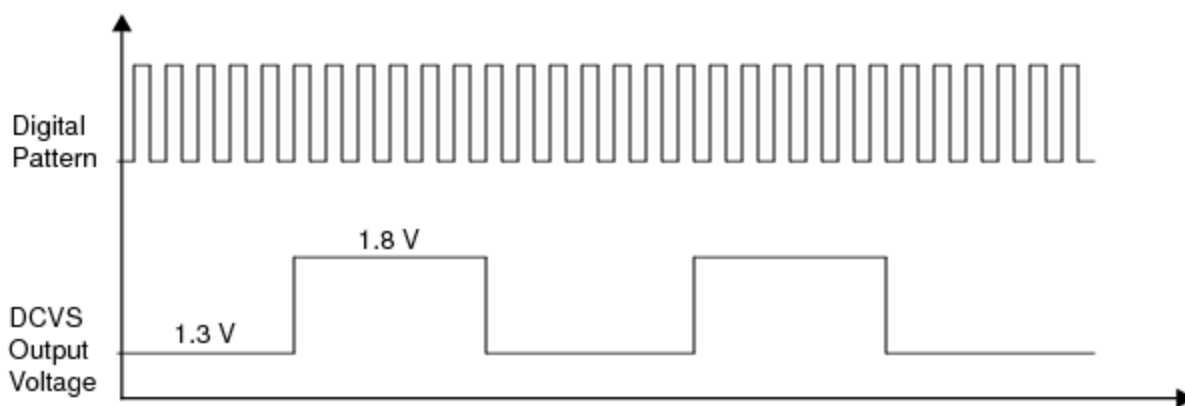
VSM Examples : Vbump Testing

Vbump Testing

A Vbump test changes the device power pin (VDD) level during a test pattern burst. You may need to conduct a Vbump test to verify that a device is making smooth power transitions as it moves between different power states. For example, you could check a device's operation at two different voltage levels (1.3V and 1.8V).

To set up a Vbump test, the VSM can change the voltage during a burst with Vmain, Valt, and parameter set programming. Note that the VSM also uses the high-speed current meter to collect the Vbump test data during pattern bursts.

To give you pattern control for Vbump testing, the VSM allows you to select one of two voltage levels during pattern bursts. Within a pattern burst, you can start and stop the current meter only as needed. The following figure models the VSM pattern control for Vbump testing.



DCVS Voltage Pattern Control for Vbump Testing

Vbump Example Using VBT PSets and a Pattern File

The following example uses PSets and a pattern file with the VS_Alt_Sel and VS_Main_Sel microcodes to “bump” the output voltage on a pin named VDD between four different values in the following order:

1.	Start at 3.0V.
----	----------------

2.	Vbump to 3.5V.
----	----------------

3.	Vbump to 3.0V.
----	----------------

4.	Vbump to 1.5V.
----	----------------

5.	Vbump to 1.75V.
----	-----------------

6.	Vbump to 1.5V.
----	----------------

7.	End at 1.5V.
----	--------------

VBT PSet Programming Function

This example uses the [AddPset](#) function to create a PSet named "my_pset_1". This function is described in the [DCVS PSet Example Function](#). For more information on programming with PSets, see [PSets](#).

Function VBumpSetup(argc As Long, argv() As String) As Long

```
' Set voltage levels and define the PSet before pattern burst
```

```
' (For example, PrePat interpose function)
```

```
Dim power_supply_pin As String
```

```
' Set the power supply pin to VDD
```

```
power_supply_pin = "VDD"
```

```
TheHdw.DCVS.Pins(power_supply_pin).Voltage.Main.Value = 3
```

```
TheHdw.DCVS.Pins(power_supply_pin).Voltage.Alt.Value = 3.5
```

```
' Create "my_pset_1" using the AddPSet() function.
```

```
Call AddPSet(power_supply_pin, _
```

```
    PSetName="my_pset_1", _
```

```
    BandwidthSetting:=0, _
```

```
    CurrentRange:=15, _
```

```
    SampleRate:=1000000, _
```

```

SampleSize:=100000, _
SourceFoldLimit:=16.5, _
MeterCurrentRange:=15, _
MeterVoltageRange:=4, _
MeterMode:=tlDCVSMeterCurrent, _
MainVoltage:=1.5, _
AltVoltage:=1.75)

```

VBumpSetup = TL_SUCCESS

End Function

Pattern File Example

The embedded comments in the following pattern file show where Vbump changes occur. For more information on programming pattern files, see [DCVS Pattern Control](#). For settle time guidelines, see [DCVS Pattern File Example](#).

```

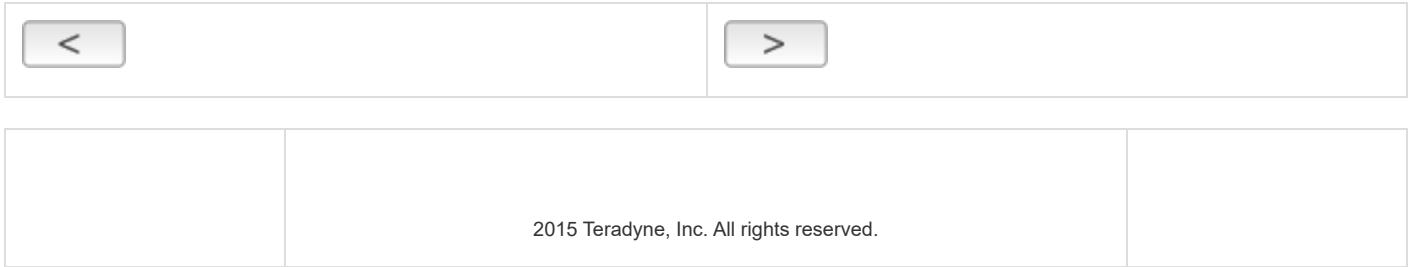
import tset ts1;

instruments = {
    VDD : DCVS;
}
srm_vector ($tset, dig1:S)
{
    // __ THIS PATTERN STARTS WITH VDD VOLTAGE at 3.0V __
    > ts1 1;
    > ts1 1;
    //... many more vectors
    > ts1 1;
    > ts1 1;
    // __ VBUMP to 3.5 V __
    (vdd : DCVS = VS_Alt_Sel)
    > ts1 1;
    // settle time repeat. See guidelines in "DCVS Pattern File Example."
    repeat 5000
    > ts1 1;
    > ts1 1;
    //... many more vectors
    > ts1 1;
    > ts1 1;
    // __ VBUMP TO 3.0 V __

```

```
(vdd : DCVS = VS_Main_Sel)
> ts1 1;
// settle time repeat. See guidelines in "DCVS Pattern File Example."
repeat 5000
> ts1 1;
> ts1 1;
//... many more vectors
> ts1 1;
// __ APPLY A PSET TO LOAD NEW VALUES OF MAIN AND ALT __
// __ THIS PSET CHANGES VDD TO 1.5 V __
(vdd : DCVS = PSet my_pset_1)
> ts1 1;
> ts1 1;
//... many more vectors
> ts1 1;
> ts1 1;
//... many more vectors
> ts1 1;
> ts1 1;
// __ VBUMP TO 1.75 V __
(vdd : DCVS = VS_Alt_Sel)
> ts1 1;
// settle time repeat. See guidelines in "DCVS Pattern File Example."
repeat 5000
> ts1 1;
> ts1 1;
//... many more vectors
> ts1 1;
> ts1 1;
// __ VBUMP TO 1.5 V __
(vdd : DCVS = VS_Main_Sel)
> ts1 1;
// settle time repeat. See guidelines in "DCVS Pattern File Example."
repeat 5000
> ts1 1;
> ts1 1;
> ts1 1;
//... many more vectors
```

}



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VSM Examples : Current Meter Setup, Strobe, and Readback Programming

Current Meter Setup, Strobe, and Readback Programming

The following example shows how to program the DCVS current meter to strobe the value of the current at a given time and store these current values in capture memory. It also shows how to read back data from the capture memory.

Dim PLD As New PinListData

Dim Pin As Variant

Dim Site As Variant

Dim PinList As String

PinList = "PinA,PinB"

With TheHdw.DCVS.Pins(PinList)

.CurrentRange.Value = 1

.Meter.Mode = tlDCVSMeterCurrent

.Meter.CurrentRange.Value = 1.25

End With

' Strobe and read back 10 samples

PLD = TheHdw.DCVS.Pins(PinList).Meter.Read(tlStrobe, 10)

For Each Site In TheExec.Sites

For Each Pin In PLD.Pins

Debug.Print "The measured value on Site(" + CStr(Site) + _
 ") of Pin (" + Pin + ") is " + CStr(PLD.Pins(Pin))

Next Pin

Next Site

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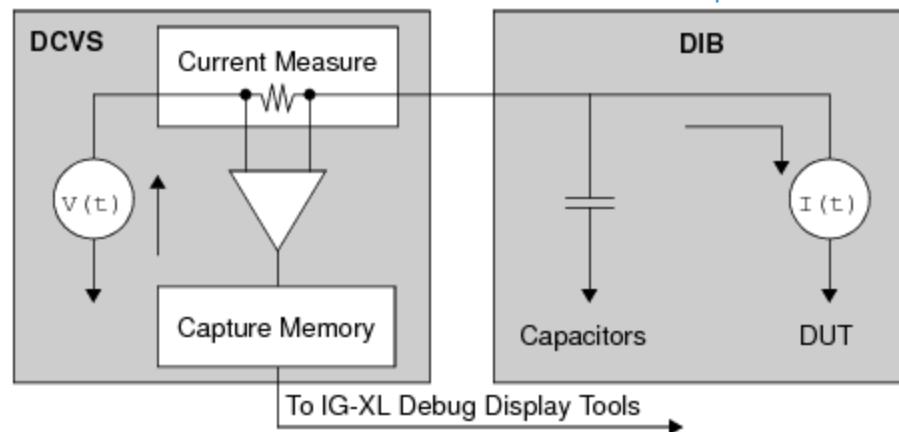


VSM Examples : Current Profiling

Current Profiling

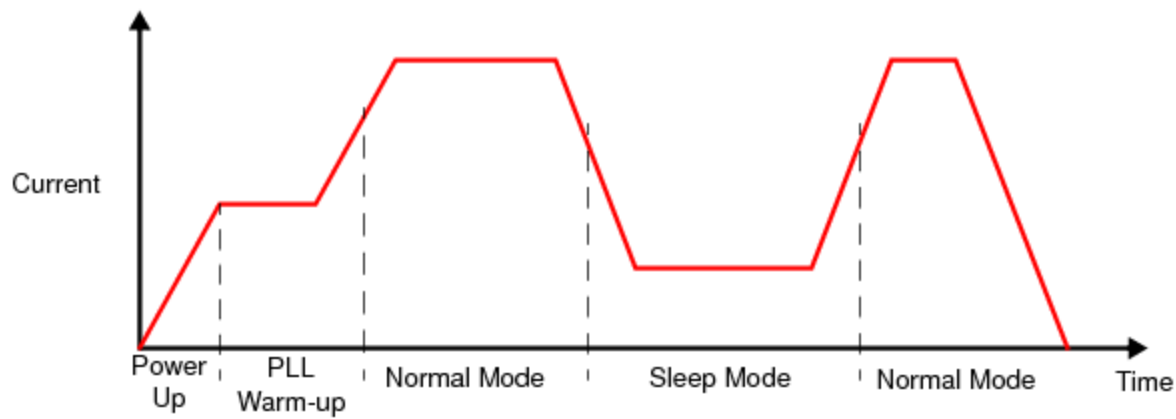
A current profile is a collection of data points of a device's current draw over time. Generally, you conduct a current profile to see what a tested part is doing as you cause it to pull more or less current. This allows you to monitor the current profile during the transition between device power states. For instance, you may want to verify that the device is drawing the expected current and is not experiencing current spikes as it moves between inactive and active modes.

The VSM also allows you to collect a current profile with a single, continuous, full-speed PatGen burst. The data is sampled at the specified frequency and stored in the on-board capture memory. Note that each VSM DCVS instrument has an independent 256K capture memory.



Current Profile Circuit Diagram

The following figure shows a hypothetical current profile generated from multiple test patterns run at speed to test a device's current requirements as it moves through different power states. The current changes as the device goes through the power-up sequence, the phase-locked loop (PLL) warm-up, and its normal and sleep modes.



Current Profile Sample

Capture Setup, Trigger, and Plot Programming

The following example shows how to capture a current or voltage waveform and plot the result.

```
Sub subProfile_Start(PlotName As String, _
    PinName As String, _
    Optional Vorl As String = "I", _
    Optional SampleRate As Double = 25000, _
    Optional SampleSize As Long = 5000)
```

```
    Dim TestCaptureLabel As String
```

```
    If SampleRate = 0 Then
        SampleRate = 1000000
    ElseIf SampleRate < 25000 Then
        SampleRate = 25000
    ElseIf SampleRate > 25000000 Then
        SampleRate = 25000000
    End If
```

```
    If SampleSize = 0 Then
        SampleSize = 100000
    ElseIf SampleSize < 1 Then
        SampleSize = 1
    ElseIf SampleSize > 256000 Then
        SampleSize = 256000
    End If
```

```
    If UCase$(Vorl) = "V" Then
        TheHdw.DCVS.Pins(PinName).Meter.Mode = tIDCVSMeterVoltage
    Else
```

```
TheHdw.DCVS.Pins(PinName).Meter.Mode = tlDCVSMeterCurrent
End If
```

```
TestCaptureLabel = PlotName
```

```
' Wait if a capture is already running.
Do While TheHdw.DCVS.Pins(PinName).Capture.IsRunning = True
Loop
```

```
TheHdw.DCVS.Pins(PinName).Capture.Signals.Add TestCaptureLabel
TheHdw.DCVS.Pins(PinName).Capture.Signals.DefaultSignal = TestCaptureLabel
```

```
With TheHdw.DCVS.Pins(PinName).Capture.Signals.Item(TestCaptureLabel)
```

```
.Reinitialize
```

```
.Mode.Mode = tlSignalModeUseLoadedValue
```

```
' When calling .LoadSettings, do not override the .Meter.Mode.
```

```
' ____ NOTE on Signals.Item(<label>).<signal_parameter>.Mode ____
```

```
' The signal parameter mode is a LoadSettings Mode for the signal.
```

```
' It determines if the parameter is set to the Signal-defined value
```

```
' or remains untouched when the .LoadSettings method is called.
```

```
' This example uses the .Mode.Mode syntax, which determines the
```

```
' behavior of the .LoadSettings function for the parameter. In the
```

```
' example syntax above, .Mode.Mode, the first .Mode is the
```

```
' DCVS.Meter.Mode (voltage or current). The second .Mode is the signal
```

```
' LoadSettings mode. All signal parameters have a LoadSettings mode.
```

```
' For instance you can also program:
```

```
' .SampleRate.Mode = tlSignalModeUseLoadedValue ' Use value already loaded to hardware.
```

```
Do not override hardware.
```

```
' .SampleSize.Mode = tlSignalModeUseLoadedValue ' Use value already loaded to hardware. Do
not override hardware.
```

```
' The default value is tlSignalModeUseValue (Use the value defined for the signal; that is,
override current hardware.)
```

```
.SampleRate = SampleRate
```

```
.SampleSize = SampleSize
```

```
.LoadSettings
```

```
.Trigger
```

```
End With
```

```
End Sub
```

```
Sub subProfile_Plot(PlotName As String, _
    PinName As String, _
    Optional DumpToFile As Boolean = False)

    Dim TestCaptureLabel As String
    Dim DSPW As New DSPWave

    ' Wait if capture is still running.
    Do While TheHdw.DCVS.Pins(PinName).Capture.IsRunning = True
    Loop

    TestCaptureLabel = PlotName

    Dim FileName As String
    Dim strMsg As String

    ' Single-pin PinListData can be read directly into DSPWave
    DSPW = TheHdw.DCVS.Pins(PinName).Capture.Signals(TestCaptureLabel).DSPWave
    If DSPW.SampleSize > 0 Then
        DSPW.Plot TestCaptureLabel
        If DumpToFile Then
            FileName = "PlotData_" & TestCaptureLabel & ".txt"
            Call PlotToFile(FileName, DSPW)
        End If
    Else
        strMsg = "subProfile_Plot() Warning: No Samples Captured (CaptureLabel=" &
TestCaptureLabel & ")"
        TheExec.Datalog.WriteComment (strMsg)
        Beep
    End If
    Set DSPW = Nothing
End Sub

Sub PlotToFile(PlotFilePathAndName As String, PlotDSPW As DSPWave)

    Dim SamplePeriod As Double
    Dim SampleSize As Long
```

```
Dim s As Long
Dim FileNo As Integer
Dim TempArray() As Double

FileNo = 7
Open PlotFilePathAndName For Output As #FileNo

SampleSize = PlotDSPW.SampleSize
SamplePeriod = (1 / PlotDSPW.SampleRate)

TempArray = PlotDSPW.data
For s = 0 To (SampleSize - 1)
    Print #FileNo, s * SamplePeriod & ", " & TempArray(s)
Next s
Erase TempArray

Close #FileNo
End Sub
```

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VSM Examples : DCVS Current Range and Limit Programming

DCVS Current Range and Limit Programming

The following example shows how to set up current limiting programming using the following DCVS “shorthand” functions.

- TheHdw.DCVS.Pins(PinList).CurrentLimit.Source.SetLimitLevels (SourceFoldLevel, SourceOverloadLevel)
- TheHdw.DCVS.Pins(PinList).CurrentLimit.Sink.SetLimitLevels (SinkFoldLevel, SinkOverloadLevel)
- TheHdw.DCVS.Pins(PinList).CurrentLimit.Source.SetLimitTimeouts (SourceFoldTimeout, SourceOverloadTimeout)
- TheHdw.DCVS.Pins(PinList).CurrentLimit.Sink.SetLimitTimeouts (SinkFoldTimeout, SinkOverloadTimeout)
- TheHdw.DCVS.Pins(PinList).SetCurrentRanges(OutputCurrentRange, MeterCurrentRange)

These functions help make DCVS programming more efficient. Each of these functions takes the place of several lines of code that would otherwise be needed to check that dependent settings (like CurrentRange and MeterCurrentRange) are compatible with the target instrument hardware. In the case of current limit programming, these functions help replace about 12 lines of VBT code for each current limit (source and sink).

Public Const UseExistingValue = -99

```

Function DCVS_SetCurrentLimits(PinList As String, _
    Optional SourceFoldLevel As Double = UseExistingValue, _
    Optional SourceFoldTimeout As Double = UseExistingValue, _
    Optional SourceOverloadLevel As Double = UseExistingValue, _
    Optional SourceOverloadTimeout As Double = UseExistingValue, _
    Optional SourceOverloadEnabled As Boolean = False, _
    -
    Optional SinkFoldLevel As Double = UseExistingValue, _
    Optional SinkFoldTimeout As Double = UseExistingValue, _
    Optional SinkOverloadLevel As Double = UseExistingValue, _
    Optional SinkOverloadTimeout As Double = UseExistingValue, _
    Optional SinkOverloadEnabled As Boolean = False _
) As Long

```

' For VS Source Current Limits

```

With TheHdw.DCVS.Pins(PinList).CurrentLimit.Source
    If SourceFoldLevel = UseExistingValue Then SourceFoldLevel = _
        .FoldLimit.Level.Value
    If SourceFoldTimeout = UseExistingValue Then SourceFoldTimeout = _
        .FoldLimit.TimeOut.Value
    If SourceOverloadLevel = UseExistingValue Then SourceOverloadLevel = _
        .OverloadLimit.Level.Value
    If SourceOverloadTimeout = UseExistingValue Then SourceOverloadTimeout = _
        .OverloadLimit.TimeOut.Value
End With
TheHdw.DCVS.Pins(PinList).CurrentLimit.Source.SetLimitLevels SourceFoldLevel, _
    SourceOverloadLevel
If TheHdw.DCVS.Pins(PinList).Gate = False Then
    TheHdw.DCVS.Pins(PinList).CurrentLimit.Source.SetLimitTimeouts _
        SourceFoldTimeout, SourceOverloadTimeout
    TheHdw.DCVS.Pins(PinList).CurrentLimit.Source.OverloadLimit.Enabled = _
        SourceOverloadEnabled
End If

```

' For VS Sink CurrentLimits

```

With TheHdw.DCVS.Pins(PinList).CurrentLimit.Sink
    If SinkFoldLevel = UseExistingValue Then SinkFoldLevel = _
        .FoldLimit.Level.Value
    If SinkFoldTimeout = UseExistingValue Then SinkFoldTimeout = _
        .FoldLimit.TimeOut.Value

```



```

    If SinkOverloadLevel = UseExistingValue Then SinkOverloadLevel = _
        .OverloadLimit.Level.Value
    If SinkOverloadTimeout = UseExistingValue Then SinkOverloadTimeout = _
        .OverloadLimit.TimeOut.Value
End With
TheHdw.DCVS.Pins(PinList).CurrentLimit.Sink.SetLimitLevels SinkFoldLevel, _
    SinkOverloadLevel
If TheHdw.DCVS.Pins(PinList).Gate = False Then
    TheHdw.DCVS.Pins(PinList).CurrentLimit.Sink.SetLimitTimeouts _
        SinkFoldTimeout, SinkOverloadTimeout
    TheHdw.DCVS.Pins(PinList).CurrentLimit.Sink.OverloadLimit.Enabled = _
        SinkOverloadEnabled
End If
End Function
Function DCVS_SetCurrentRanges(PinList As String, _
    Optional OutputCurrentRange As Double = UseExistingValue, _
    Optional MeterCurrentRange As Double = UseExistingValue _
) As Long
If OutputCurrentRange = UseExistingValue Then OutputCurrentRange = _
    TheHdw.DCVS.Pins(PinList).CurrentRange.Value
If MeterCurrentRange = UseExistingValue Then MeterCurrentRange = _
    TheHdw.DCVS.Pins(PinList).Meter.CurrentRange.Value
TheHdw.DCVS.Pins(PinList).SetCurrentRanges OutputCurrentRange, MeterCurrentRange
End Function
```

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DCVS and VSM Board Debug Displays

DCVS and VSM Board Debug Displays

This section describes both the DCVS instrument and the VSM board debug displays:

- [DCVS Debug Display](#)
- [VSM Board Debug Display](#)

For more information on using debug displays, see the following:

- [Using Debug Displays in TDE](#)
- [Producing Code from Debug Displays](#)

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DCVS and VSM Board Debug Displays : DCVS Debug Display

DCVS Debug Display

The interactive DCVS debug display allows you to troubleshoot DCVS settings. You can change any values on the display and run your test program to see the effects of the change.

This section includes the following topics:

- Starting the DCVS Debug Display
- DCVS Debug Display Interface
- Setting the Output Current Range and Meter Ranges

Starting the DCVS Debug Display

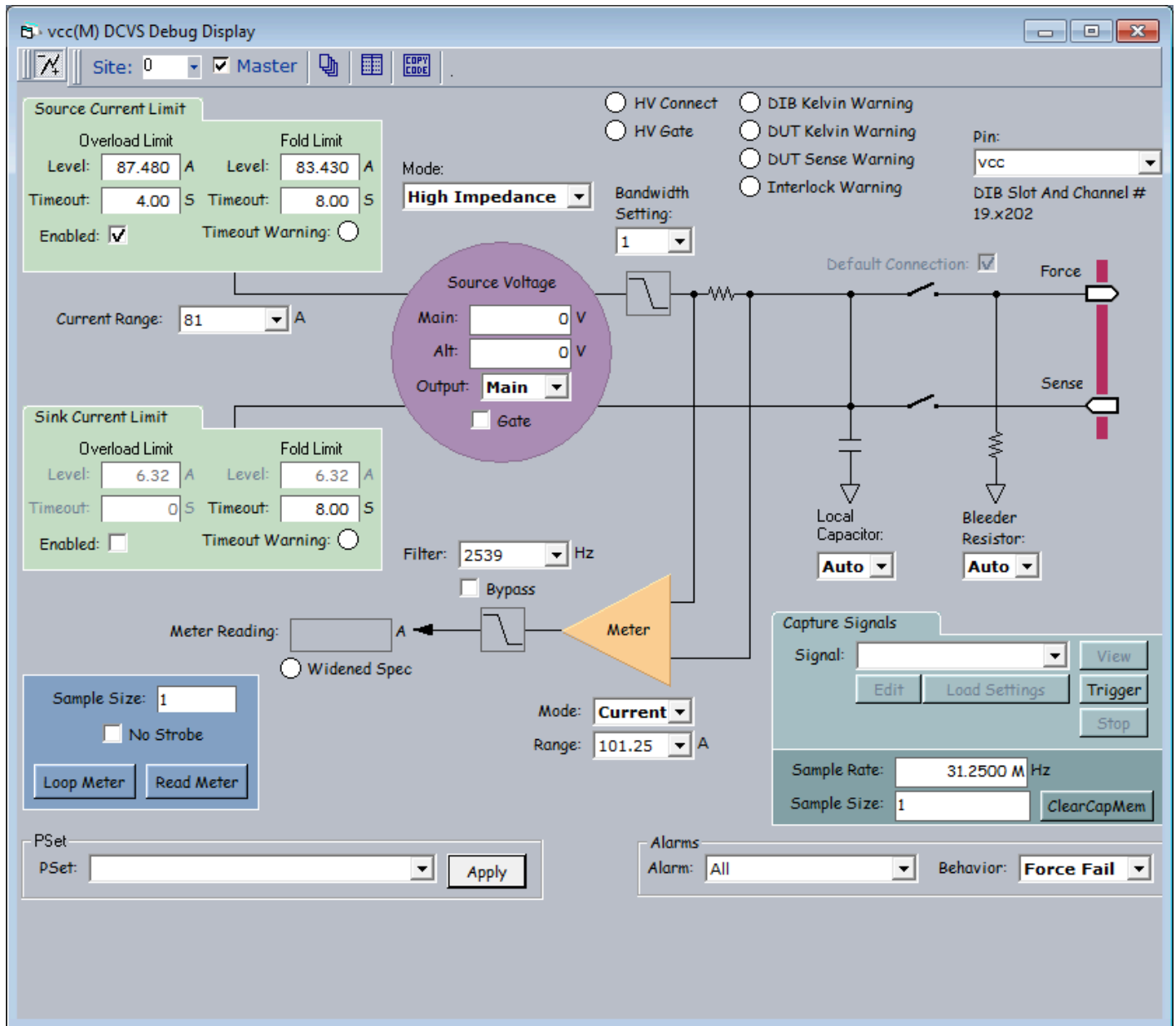
To launch the DCVS debug display for VSM, first validate a VSM test program. Then, on the IG-XL Tools tab, in the Launch group, select Debug Displays>DCVS. This opens the display for active sites only (see figure below).

If you start a debug display before the job has been validated, an error appears in the display saying that the program has not been validated. If you open a display for an instrument that is in the configuration file but not in the channel map, the display shows a message that no instruments have been found.

Once the job is validated, the display shows the current hardware settings. When you run to a breakpoint in the flow or in the VBT code, or as you step through the flow or code, IG-XL updates the display to show any changes to the settings.

For information on other methods for starting debug displays, see Starting Debug Displays in the Test Debug Environment (TDE) section.

For information on fields used in the DCVS debug display, see DCVS Debug Display Interface.



DCVS Debug Display with VSM Values

DCVS Debug Display Interface

This section describes the toolbar menus and buttons and the fields used in the DCVS debug display. To change values in the debug display, either type the new values in the text box or select the new value from the menu.

Note:

To view the ranges or available options for fields that do not have drop-down menus, place your pointer over the field and the minimum and maximum values appear in a pop-up window. These ranges vary depending on which VS board you have assigned to the DCVS channel and how you have programmed it.

The table lists DCVS Debug Display parameters, buttons, and fields used by the VSM.

DCVS Debug Display Fields for VSM

Menu, Button, or Field	Action
Source Current Limit	Use the following fields to view or set your source current limiting hardware: • Overload Limit: The overload current level limit in the Level field and the timeout period in the Timeout field. • Fold Limit: The fold current level limit in the Level field and the timeout period in the Timeout field. • Enabled: Checkbox enables the source overload current limiting. • Timeout Warning: Red circle indicates the fold timeout has expired. • Fold Level Changed: Not supported on VSM. See Setting the Current Limit.
Sink Current Limit	Use the following fields to view or set your source current limiting hardware: • Overload Limit: The overload current level limit in the Level field and the timeout period in the Timeout field (disabled for VSM). • Fold Limit: The fold current level limit in the Level field (disabled for VSM) and the timeout period in the Timeout field. • Enabled: Checkbox enables the sink overload current limiting. • Timeout Warning: Red circle indicates the fold timeout has expired. See Setting the Current Limit.
Source Voltage	Use the following fields to set up your output voltage:

- **Main:** The main source voltage value in volts.
- **Alt:** The alternate source voltage value in volts.
- **Output:** The output for the source voltage, either Main or Alt.
- **Gate:** The DCVS is gated on (checked) or off (empty).

Mode

The DCVS operation mode. Options:

- **Force Voltage** (On)
- **High Impedance** (Off)

Pin

The pin you are observing. The DIB slot and channel number you are currently viewing are shown below this field.

Bandwidth

The compensation mode. Options: 0 through 4. For specific DIB design guidelines on choosing a compensation setting, see [Setting the Output Compensation Bandwidth](#).

Current Range

The output current range. See [Setting the Output Current Range and Meter Ranges](#).

Default Connection

You cannot change this connection on VSM. Checkbox determines how the force and sense connections are opened and closed. When checked, these connections are opened and closed together. When unchecked, force and sense connections are controlled independently.

Local Capacitor

The connection state of the local output capacitor. When the DCVS is disconnected from the DIB, the default capacitor keeps the DCVS voltage output from oscillating. For VSM, the local capacitor is switched on to stabilize the power supply in Auto mode. Options:

- **On**
- **Off** (Not supported on VSM)

- [Auto](#)

See [Local Output Capacitor, Bleeder Resistor, and Rapid Discharge](#).

Bleeder Resistor

The connection state of the bleeder resistor. When the DCVS is disconnected from the DIB, and the bleeder resistor helps quickly discharge DIB capacitance. Options:

- [On](#)

- [Off](#)

- [Auto](#)

See [Local Output Capacitor, Bleeder Resistor, and Rapid Discharge](#).

Meter

The DCVS meter settings, including:

- [Mode](#): The meter mode, either Current or Voltage.
- [Range](#): The meter range. See [Setting the Output Current Range and Meter Ranges](#).
- [Filter](#): The meter bandwidth frequency of the low-pass filter (for viewing and programming).
- [Bypass](#): Checking this box disables the filtering, allowing to view a higher frequency signal with more stability (not supported by all DCVS filters).
- [Meter Reading](#): The measured voltage or current value.
- [Sample Size](#): The number of samples strobed.
- [No Strobe](#): Checking this box disables the strobe function.

- [Loop Meter](#): Clicking this button opens the Meter Loop Tool and causes the meter to loop through multiple strobe sets. See [Meter Loop Tool](#).
- [Read Meter](#): Clicking this button prints the average of the strobed samples.

PSet

A list of defined PSets for selected pin site. When no PSet is defined, the list is empty. When a PSet is selected, the PSet is applied to instrument hardware immediately. The box is updated to display any PSet that is applied from VBT.

Alarms

The alarm type and behavior for the selected alarm.

- [Alarm](#): The alarm type.
- [Behavior](#): The action that the alarm takes when triggered.

See [Kelvin Sensing and Alarms Subsystem](#) and [Setting Alarms](#).

Capture Signals

The list of capture signals defined in VBT language, including the following capture memory signal settings:

- [Signal](#): A capture signal tag listed in the menu.
- [Load Settings](#): Clicking this button loads the capture signal tag's settings.
- [Edit](#): Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. See [PSet and Signals Editor](#).
- [View](#): Clicking this button shows the selected capture signal tag's waveform.
- [Trigger](#): Clicking this button triggers new capture signal with the current hardware setup.

- [Stop](#): Clicking this button stops the current capture, if one is running.
- [Sample Rate](#): The capture rate currently set in hardware.
- [Sample Size](#): The capture sample size currently set in hardware.
- [ClearCapMem](#): Clicking this button clears both the signal (capture) and meter memories for channels behind the specified pin or pins.

DIB Kelvin Warning	A red circle indicates that the DIB Kelvin threshold has been exceeded.
DUT Kelvin Warning	A red circle indicates that the DUT Kelvin threshold has been exceeded.
DUT Sense Warning	A red circle indicates that the DUT Sense threshold has been exceeded.
Interlock Warning	A red circle indicates that either the DIB is not seated correctly on the test head or a connection problem exists in the DCVS interlock circuit.
HV Gate Warning	A red circle indicates that the high voltage gate (HV_GATE) safety signal from the support board is not connected properly to DIB ground. This usually means that the DIB is either not properly seated or designed incorrectly.
HV Connect Warning	A red circle indicates that the high voltage connect (HV_CONNECT) safety signal from the support board is not connected properly to DIB ground. This usually means that the DIB is either not properly seated or designed incorrectly.
Widened Spec Warning	A red circle indicates that the specification has widened beyond its limits. This means the board is outside of its optimal operating temperature; it can still operate, but it cannot meet its specification.

Setting the Output Current Range and Meter Ranges

The available output current ranges and corresponding meter ranges depend on whether you have merged your DCVS channels or left them independent. For a nonmerged DCVS channel, the only

valid option is 81A. However, for merged DCVS channels, you have the choice between using the 81A option or the merged range options (162A, 243A, 324A, 405A, or 486A). For more information, see [VSM Output Current and Current Measurement Ranges](#).

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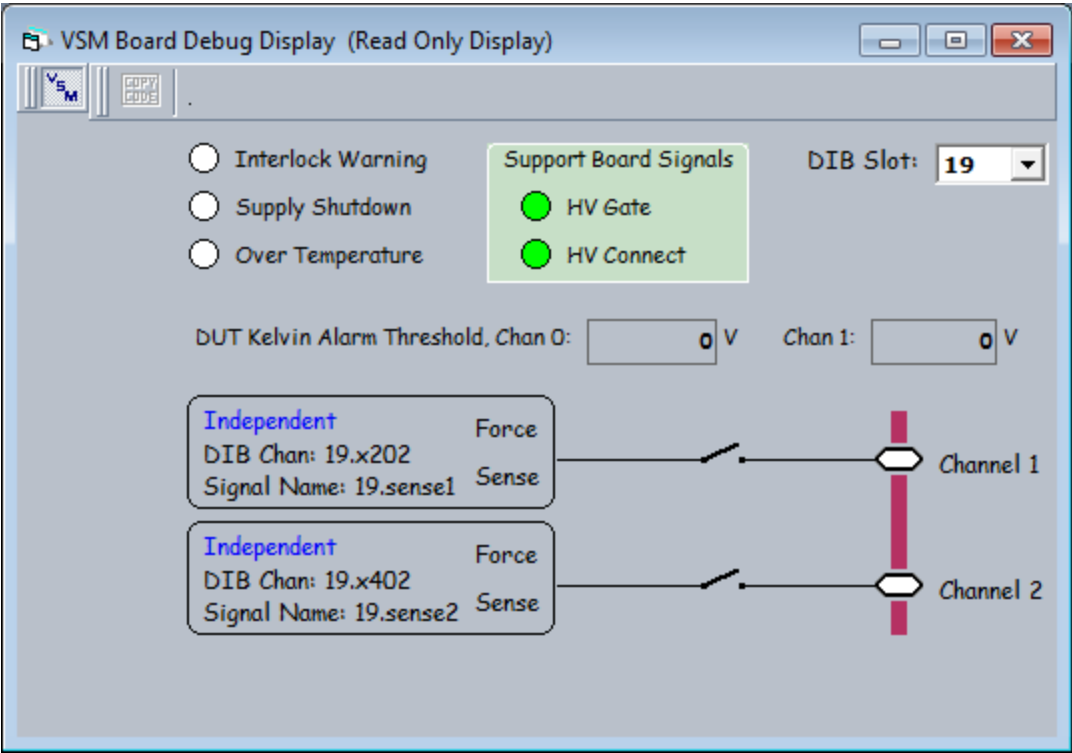
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DCVS and VSM Board Debug Displays : VSM Board Debug Display

VSM Board Debug Display

This read-only debug display allows you to see how DCVS instruments are connected to the DIB. It also shows merged DCVS instruments.

To launch the VSM Board Debug Display, first validate a VSM test program. Then, on the IG-XL Tools tab, in the Launch group, select Subsystem>VSM Board. This brings up the VSM debug display shown in the figure below.



VSM Board Debug Display

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VSM DCVS Programming

VSM DCVS Programming

This section describes programming of the VSM board’s DCVS instruments.
The following topics are available:

- | | |
|---|---|
| • | Simulated Configuration Files |
| • | Programming Procedure Overview |
| • | Before You Begin |
| • | DCVS DataTool Programming |
| • | DCVS VBT Programming |
| • | DCVS Pattern Control |
| • | DCVS Test Instance Programming |
| • | Using VSM in Concurrent Testing |

 Note:

The VSM instrument provides offline simulation capabilities. Since each board and channel is modelled independently, you can write and execute jobs offline. IG-XL uses a VSM behavioral model when executing language statements, but the VSM simulation does not model device

behavior. When you take simulated measurements, IG-XL returns the voltage value settings and a current value of 0A. IG-XL does not simulate pattern-controlled operations.
For more information, see:

- [DCVS \(VBT Syntax Reference\)](#)
- [VSM Examples](#)

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VSM DCVS Programming : Simulated Configuration Files

Simulated Configuration Files

You can use the VSM offline by forcing IG-XL to map the VSM. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the VSM board in slot 21:

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[.subslot]  Type              idprom (type serial rev company)

4.0              HSD-M              979-219-00 00000000 0000-A 0000
                  HSD-M              956-361-00 00000000 0000-A 0000

12.0             HexVS              974-294-00 00000000 0000-A 0000
                  HexVS              956-388-00 00000000 0000-A 0000

21.0             VSM                974-296-00 00000000 0000-A 0000
```

VSM

956-361-00 0000000 0000-A 0000

24.0

SupportBoard

974-220-12 0000000 0000-A 0000

SupportBoard

956-354-00 0000000 0000-A 0000

62.0

DistributionBoard

956-355-00 0000000 0000-A 0000

DistributionBoard

956-355-00 0000000 0000-A 0000

</TEST_HEAD TH 1>

To use the VSM offline, the `SimulatedConfig.txt` file must include every board needed in the IG-XL workbook. The `SimulatedConfig.txt` file represents the state of the offline tester. To use the VSM online, you need not edit any files. IG-XL automatically maps all boards in the tester. For more information, see [Tester Configuration](#).

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VSM DCVS Programming : Programming Procedure Overview

Programming Procedure Overview

To develop a test program that uses the DCVS instrument, follow this basic procedure:

1. Determine device power supply requirements (Device Power Supply Requirements).

a. Number of power pins

b. Output voltage and current ranges for each power pin

c. Maximum current limiting requirements

2. Consider the following features specific to the DCVS instrument (DCVS Instrument Considerations):

a. Instrument board-specific DIB design considerations such as thermal control unit connections for output current monitoring.

b. Output power compliance

c. Merging DCVS instrument supplies

3. Create a Pin Map sheet (Programming the Pin Map).

4. [Create a Channel Map sheet \(Programming the Channel Map\).](#)
5. [Create a Pin Levels sheet \(Programming Pin Levels\).](#)
6. [Set the maximum current limits using VBT code \(Setting the Current Limit\).](#)
7. [Set the compensation mode using VBT code \(Setting the Output Compensation Bandwidth\).](#)
8. [Set the alarms using VBT code \(Setting Alarms\).](#)
9. [Set the DCVS meter to measure current using VBT code \(Setting the DCVS Current Meter\).](#)
10. [Create DCVS test patterns to use in your test instances \(DCVS Pattern Control\).](#)
11. [Create test instances in the Test Instances sheet \(DCVS Test Instance Programming\).](#)
12. [Add test instances to the Flow Table sheet.](#)
13. [Run the test program, debug run-time errors or alarms, and optimize for run-time performance.](#)

[VBT syntax](#) is described in the [DCVS](#) section of the [VBT Syntax Reference](#).

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VSM DCVS Programming : Before You Begin

Before You Begin

Before writing a DCVS test program, you must determine the power requirements for your device application, including the number of power pins, the output voltage and current ranges for each power pin, and the maximum current limiting requirements. Moreover, you must understand the features of the DCVS instrument and how they affect your test program development. This section covers both of these topics:

- | | |
|---|----------------------------------|
| • | Device Power Supply Requirements |
| • | DCVS Instrument Considerations |

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VSM DCVS Programming : Before You Begin : Device Power Supply Requirements

Device Power Supply Requirements

Review the device datasheet to determine the device’s power pin requirements. Note, however, that during testing, the device power requirements are likely to exceed the standard functional values listed in the datasheet. When possible, therefore, consult the device product engineer to get an idea of the predicted device power requirements during testing. In general, new generation digital VLSI devices tend to draw more power supply current during testing than previous generation devices.

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VSM DCVS Programming : Before You Begin : DCVS Instrument Considerations

DCVS Instrument Considerations

Consider the following before creating a DCVS test program.

DCVS Instrument DIB Design Guidelines

For information on DIB design guidelines, including DIB signals and connections and wiring guidelines, see [VSM DIB Design Guidelines](#).

Output Power Compliance

Make sure you understand the output power specifications of the VSM board and the DCVS instrument before setting up your test program. For more information, see [Output Power](#).

Merging DCVS Instrument Supplies on VSM

For the most part, programming merged DCVS instruments is similar to programming a single DCVS. The primary difference is that the test program channel map indicates that the DCVS instrument uses either a DCVSMerged2 configuration or the combined pins notation (:c). When merging two channels, you can use either a DCVSMerged2 configuration or the :c notation. You must use the :c notation for merging three or more channels.

Another significant difference is that merged supplies have additional output current ranges. Beyond that, the only other difference is that the maximum source and sink current ranges and limit settings have increased by a factor proportional to the number of merged DCVS instruments. Voltage ranges, timer ranges, capture sample rates, and most other instrument parameters have the same settings as the nonmerged leader channel.

Note:

When merging VSM channels, the first two merged channels must be on the same board. The third, fourth, fifth, and sixth channels can be added from any other boards in any combination. That is, four channels do not have to come from two boards.

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[VSM DCVS Programming](#) : DCVS DataTool Programming

[DCVS DataTool Programming](#)

This section describes how to program several DataTool worksheets to handle basic DCVS parameters. It includes the following topics:

- | | |
|---|---|
| • | Programming the Pin Map |
| • | Programming the Channel Map |
| • | Programming Pin Levels |

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[VSM DCVS Programming](#) : [DCVS DataTool Programming](#) : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. Select the Power pin type for each device pin name connected to a DCVS instrument.
For more information, see [Pin Map Sheet](#) in the DataTool section.

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VSM DCVS Programming : DCVS DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For the DCVS instrument, you must select a channel type from the Type column menu for each pin or pin group. Valid DCVS channel types are:

DCVS Instrument Channel Type Options	
Option	Explanation
DCVS	Standard connection to a single DCVS instrument on one VSM board. Provides up to 81A of output current.
DCVSMerged2	Indicates a DIB wiring connection that merges two DCVS instruments on one VSM board to provide up to 162A of output current. Supports board-to-board merging. When merging VSM channels, the first two merged channels must be on the same board. The third, fourth, fifth and sixth channels can be added from any other boards in any combination. That is, four channels do not have to come from two boards.
DCVSMerged4	Not supported on VSM.
DCVSMerged6	Not supported on VSM.

When assigning channels for a specific site, use the following channel map signal name format:
<slot_number>.sense<leader_vs_number>
where:

- <slot_number>DIB slot where the VSM board is installed
- <leader_vs_number>DCVS leader channel number. For the DCVSMerged2 option, this should be sense1. When the pin name is followed by the combined pins notation (:c), then this can be any channel number.

For on-board merging of two channels, you can use either the DCVSMerged2 or the DCVS channel type option with the combined pins notation (:c). You must use the :c notation for merging three or more channels. The following figure shows examples of two equivalent channel map setups for VSM on-board merging.

Channel Map

DIB ID: View Mode:

Device Under Test		Tester Channel	
Pin Name	Package Pin	Type	Site 0
vdd		DCVSMerged2	21.sense1

Channel Map

DIB ID: View Mode:

Device Under Test		Tester Channel	
Pin Name	Package Pin	Type	Site 0
vdd		DCVS	21.sense1
vdd:c		DCVS	21.sense2

On-Board Merging Channel Map Setups

For board-to-board merging, you can use the either the DCVS or DCVSMerged2 channel type option with the combined pins notation (:c). The following figure shows examples of two equivalent channel map setups for VSM board-to-board merging.

✔ **Note:**

DCVSMerged4 and DCVSMerged6 channel type options do not support board-to-board merging.

Channel Map

DIB ID: View Mode:

Device Under Test		Tester Channel	
Pin Name	Package Pin	Type	Site 0
vdd		DCVSMerged2	21.sense1
vdd:c		DCVSMerged2	22.sense1

Channel Map

DIB ID: View Mode:

Device Under Test		Tester Channel	
Pin Name	Package Pin	Type	Site 0
vdd		DCVS	21.sense1
vdd:c		DCVS	21.sense2
vdd:c		DCVS	22.sense1
vdd:c		DCVS	22.sense2

Board-to-Board Merging Channel Map Setups

For more information, see Channel Map Sheet in the DataTool section.

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VSM DCVS Programming : DCVS DataTool Programming : Programming Pin Levels

Programming Pin Levels

A Pin Levels sheet defines values for output voltages to be applied to the DUT. Once you insert this sheet into a test, you can program the Power Supply Test Instance Editor, which uses values assigned in this sheet.

The DCVS instrument supports incremental levels, which allows you to change levels for an individual pin or a pin group that has already been defined on the Pin Levels sheet. IG-XL applies the last set of specified levels for a pin or pin group when adding a new pin or pin group, applying and sequencing levels, characterizing pins, and determining specification dependencies. With this flexible programming model, you can reassign new pin levels to existing pins on the sheet while preserving previous pin level setups. You can always revert to previous levels settings and behavior by removing the most recent pin level change on the sheet. For more information, see Incremental Levels Mode.

For the DCVS instrument, you must assign a pin levels parameter for each pin name or group. The following table shows the DCVS instrument pin level parameter options.

DCVS Instrument Pin Levels Parameter Options	
Option	Explanation
VMain	Main voltage power supply. Used to supply voltage to the DUT through the main DAC. Range: 0.0V to 4.0V. Replaces the Vps parameter in IG-XL V6.10.00 and later.
VAIt	Alternate voltage power supply. Used to supply voltage to the DUT through the alternate DAC. Range: 0.0V to 4.0V.
IFoldLevel	The fold current limit level. Ranges: N x (81A to 486A), where N is the number of merged supplies. Default: N x 83.43A.

	Replaces the <code>Isc</code> parameter in IG-XL V6.10.00 and later. See Setting the Current Limit.
<code>IOverloadLevel</code>	The overload current limit level. Optional. Ranges: $N \times (81A \text{ to } 486A)$, where N is the number of merged supplies. Default: $N \times 87.48A$. Replaces the <code>Isc</code> parameter in IG-XL V6.10.00 and later. See Setting the Current Limit.
<code>Tdelay</code>	Time delay. Used to set an additional delay time when applying this set of values, which allows for more settling time when a larger voltage change is requested. <code>Tdelay</code> is not used during the power-down sequence. It is only used after powering-up all power pins of the same sequence number. If you place more than one DCVS pin within the same sequence group on the Pin Levels sheet, then IG-XL applies power to all of those pins at once and uses the maximum of all the <code>Tdelay</code> values for these pins. IG-XL automatically enforces a 1.75ms standard delay time on a VSM channel for a Pin Levels sheet change. If this time is insufficient, you can program the <code>Tdelay</code> parameter to add to this standard delay time. For example, if you program <code>Tdelay</code> as 3ms, the actual delay time will be 4.75ms. Note also that IG-XL uses only the longest programmed delay time for channels of the same sequence number because it executes time delays from all instrument boards in parallel.
<code>Vps</code>	Voltage power supply. Used to supply voltage to the DUT. Range: 0.0V to 4.0V. This parameter is used in IG-XL V6.00.08 and earlier. Use <code>VMain</code> and <code>VAlt</code> to set voltage levels in V6.10.00 and later.
<code>Isc</code>	Current source clamp. Used to set the current limits (specifically, the source fold limit level) for the DCVS. Two current limiting modes: single and dual. Ranges: $N \times (81A \text{ to } 486A)$, where N is the number of merged supplies. Default: $N \times 81A$. This parameter is used in IG-XL V6.00.08 and earlier. Use <code>IFoldLimit</code> and <code>IOverloadLimit</code> to set current limit levels in V6.10.00 and later. See Setting the Current Limit.

For more information, see [Pin Levels Sheet](#) in the DataTool section.

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[VSM DCVS Programming](#) : DCVS VBT Programming

DCVS VBT Programming

Several basic DCVS programming tasks require that you use VBT syntax.

- | | |
|---|---|
| • | Setting the Current Limit |
| • | Setting the Output Compensation Bandwidth |
| • | Setting Alarms |
| • | Setting the DCVS Current Meter |

For more information, see [DCVS](#) in the VBT Syntax Reference.

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VSM DCVS Programming : DCVS VBT Programming : Setting the Current Limit

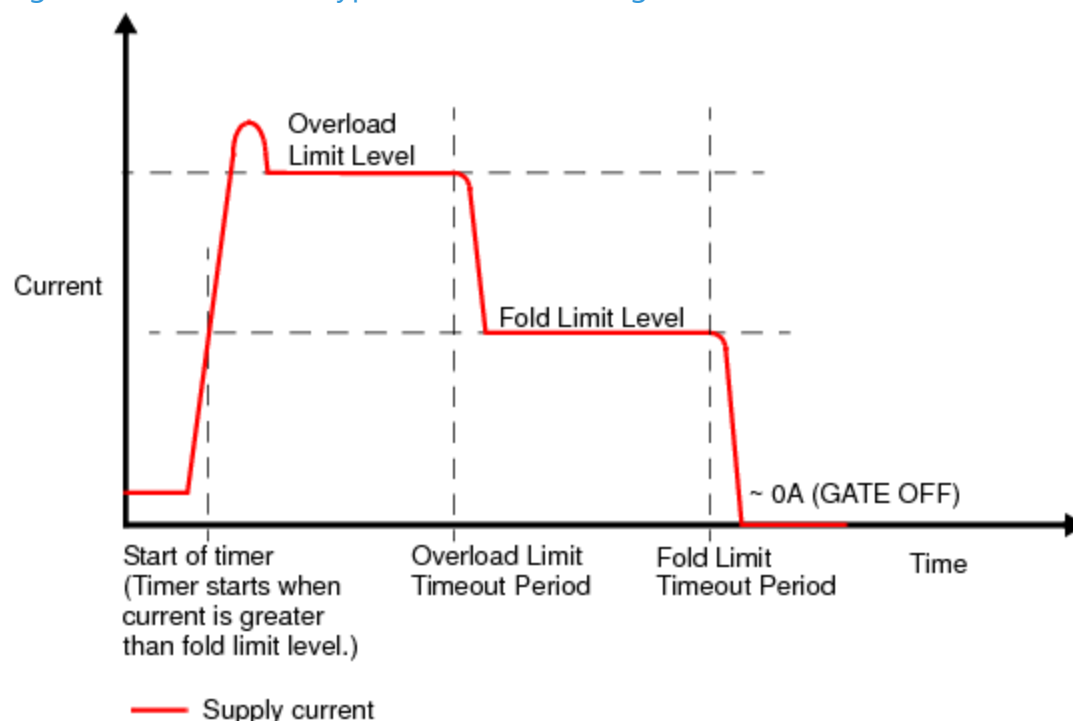
Setting the Current Limit

The DCVS instrument has two current limiting modes: single-level and dual-level. The default setting is dual-level current limiting where overload limiting is enabled. You must decide which one is best to use based on your device's power supply current limiting requirements.

For an example of how to program current range and limits using DCVS "shorthand" functions, see [DCVS Current Range and Limit Programming](#).

Dual-Level Current Limiting (Overload Limit Enabled)

VSM defaults to this type of current limiting. The hardware limits the current at two different programmable values with two different user-defined current limit timeout values. The following figure shows how this type of current limiting works.



Dual-Level Current Limiting

When the output current exceeds the fold limit level, a timer starts. To activate dual-level mode, you must enable overload limiting. In dual-level mode, the current is not immediately limited to the fold limit level. Instead, the output current is allowed to exceed the fold limit level until the overload limit timeout period expires. However, during this time the current is limited to the overload limit level. Once the overload timeout has elapsed, the output current is limited to the fold limit level.

If the current drops below the fold limit level, the current limit timer is reset. Note that the VSM hardware has a 10s period before it detects the need to clamp. If the timer is not reset and the fold limit timeout elapses, the hardware forces the DCVS output to 0A, and the VSM gates off.

Optionally, you can program the output to stay gated on for as long as possible after the fold limit timeout period has elapsed. To do this, use this VBT syntax:

```
TheHdw.DCVS.Pins.CurrentLimit.Source.FoldLimit.Behavior =  
    tIDCVSCurrentLimitBehaviorDoNotGateOff
```

To change any of the DCVS overload or fold limit parameters, use either a DCVS Power Supply VBT Test or VBT syntax. You can use the DataTool Pin Levels sheet to set the source current limit defaults.

The following example shows how to program the dual-level source current limit using VBT syntax:

With TheHdw.DCVS.Pins(PinList)

```
.Gate = False ' You MUST turn off gate to:  
    ' a) Enable or Disable the Overload Limiting  
    ' OR  
    ' b) Change Fold or Overload Limit Timeout values  
    ' otherwise a run-time error will occur.
```

With .CurrentLimit.Source

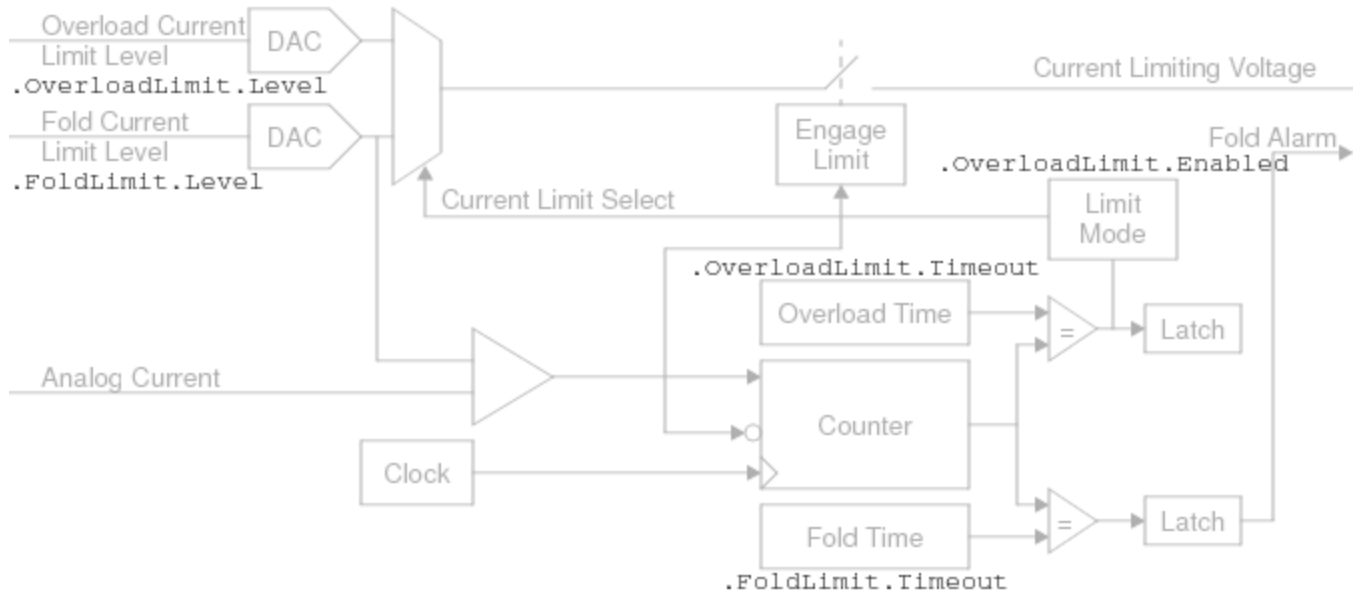
```
.OverloadLimit.Enabled = True  
.FoldLimit.Level.Value = (0.65 * .FoldLimit.Level.Max)  
.FoldLimit.Timeout.Value = 0.010  
.OverloadLimit.Level.Value = .OverloadLimit.Level.Max  
.OverloadLimit.Timeout.Value = 0.001
```

End With

End With

The following figure shows how the VBT code programs the hardware components.

Starting VBT Driver Interface: TheHdw.DCVS.Pins().CurrentLimit.Source



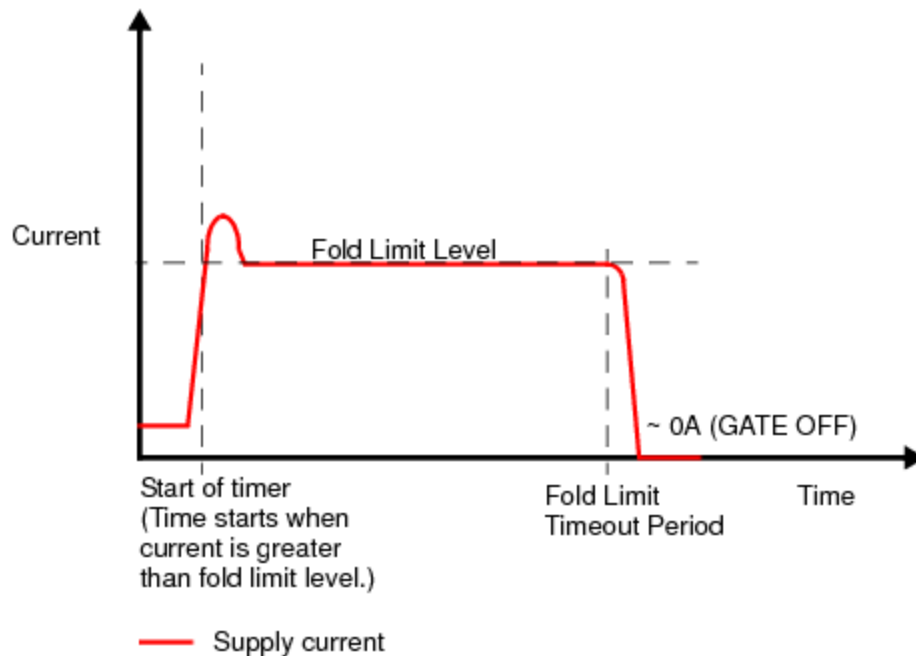
Notes:

1. Overload Limit Level > Fold Limit Level + Minimum Limit Level Difference
2. Overload Time < Fold Time (Cannot change timeout value while Gate = True)
3. Programmed Fold Limit Level is subject to output and adjacent pair power compliance restrictions.
4. Minimum Limit Level Difference is specified in the instrument board specification.

DCVS Instrument: Programming the Source/Sink Current Limiting Subsystem

Single-Level Current Limiting (Overload Limit Disabled)

The hardware limits the source or sink current to a programmed value until the user-defined fold time expires or the current drops below the limit level value. The following figure shows this type of current limiting.



Single-Level Current Limiting

If the current is greater than the fold limit level, the fold limit timer starts, and the current is immediately limited to the fold limit level. Note that the VSM hardware has a 10s period before it detects the need to clamp. The timer resets if the current drops below the fold limit level. If the

timer is not reset and the fold limit timeout period expires, however, the hardware forces the DCVS output to 0A, and the VSM gates off.

Optionally, you can program the output to stay gated on for as long as possible after the fold limit timeout period has elapsed. To do this, use this VBT syntax:

```
TheHdw.DCVS.Pins.CurrentLimit.Source.FoldLimit.Behavior =  
    tIDCVSCurrentLimitBehaviorDoNotGateOff
```

After determining the desired source and sink current limit types, you can program the overload and fold limit levels and timeout values. To activate single-level mode, you must disable overload limiting.

Use the DCVS Pin Levels sheet to set up the source current limit defaults:

•	Overload limiting: Disabled
•	Fold limit level: Isc level specified in the Pin Levels sheet
•	Fold limit timeout value: 0.010 seconds
•	Fold limit behavior: Gate Off

This VBT syntax example shows how to program the single-level source current limit:

With TheHdw.DCVS.Pins(PinList)

```
.Gate = False ' You MUST turn off gate to:  
    ' a) Enable or Disable the Overload Limiting  
    ' OR  
    ' b) Change Fold or Overload Limit Timeout values  
    ' otherwise a run-time error will occur.
```

With .CurrentLimit.Source

```
.OverloadLimit.Enabled = False  
.FoldLimit.Level.Value = (0.65 * .FoldLimit.Level.Max)  
.FoldLimit.Timeout.Value = 0.010  
.FoldLimit.Behavior = tIDCVSCurrentLimitBehaviorGateOff
```

End With

End With

DCVS Instrument: Programming the Source/Sink Current Limiting Subsystem shows how the VBT code programs the hardware components.

VSM Default Current Limit Settings

The default current limit settings when IG-XL is started are as follows:

Source Overload Limit Enabled	False
Source Overload Limit Level	N*87.48A (This value is 108% of 81A.)
Source Overload Limit Timeout	4s (Minimum Overload Limit Timeout Value)
Source Fold Limit Level	N*83.43A (This value is 103% of 81A.)
Source Fold Limit Timeout	8s
Source Fold Limit Behavior	Gate Off

where N = Number of merged VS outputs

The Pin Levels sheet parameter `Isc` is used to set the Source Fold Limit Level when the levels worksheet levels are applied (for example, `TheHdw.PinLevels.ApplyPower`). All other source current limit parameters (source overload level, timeout, and enable) and all sink parameters are not changed. These can only be changed using VBT syntax or a DCVS Power Supply VBT Test.

VSM Current Limit Programming Guidelines

Follow the guidelines listed below when programming current limits:

- Program the overload timeout value less than the fold time value. You will encounter run-time errors if you do not follow this rule.
- Keep the overload level greater than the fold level plus the minimum limit level difference in any range. See the instrument board’s specification document for the minimum limit level difference specification. You will encounter run-time errors if you do not follow this rule.
- Separate the overload and fold levels by 5% of the output current range. For example, if the fold level is 45% of the output range, then the overload must be 50% of the output range. Since you set these values in Amps, a pop-up error message provides guidance when programming these values.

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VSM DCVS Programming : DCVS VBT Programming : Setting the Output Compensation Bandwidth

Setting the Output Compensation Bandwidth

The bandwidth parameter allows the DCVS instrument to provide optimal performance with various output current range modes and DIB capacitance. These settings allow you to select the desired performance with your DIB and bypass capacitors.

Use the following syntax to specify the selection of compensation bandwidth.

TheHdw.DCVS.Pins.BandwidthSetting.Value = 0 | 1 | 2 | 3 | 4

To obtain a fast dynamic performance, you must adjust a DCVS instrument's internal compensation bandwidth for the output current range and DIB capacitance. By changing the bandwidth, the response can be optimized for minimum droop or stable operation with smaller DIB capacitance. By selecting the proper bandwidth, a DCVS can accommodate a wide range of DIB capacitance. Note that changes to the output current range affect the performance of a given bandwidth. The DCVS meets or exceeds its dynamic performance specifications if the power supply's sense lines are at the DIB capacitance and if you meet the minimum design specifications for the DIB and its capacitance.

Note that DC sensing on the die while AC sensing at the DIB capacitance compromises the power supply's droop at the die. Reducing the impedance between the DIB capacitance and the DUT (die) minimizes this effect.

Note:

Not all bandwidth settings can be used effectively across all capacitance ranges. Generally, the lower bandwidth settings (2 and 3) accommodate lower capacitance values, and the higher bandwidth settings (0 and 1) accommodate higher capacitance values.

General Compensation Guidelines

Use the following guidelines to help you choose the compensation setting:

- If you use the minimum recommended DIB capacitance value, then use compensation bandwidth setting 0 to meet the supply's droop specification.
- If you have less DIB capacitance, then compensation bandwidth setting 1, 2, or 3 provides stability, but with increased droop.
- Compensation bandwidth setting 4 provides about the same droop as bandwidth setting 1, but with six to ten times longer settle time. This setting is useful for lower current ranges with very large DIB capacitance.
- If you encounter stability problems, use the following troubleshooting techniques in the order shown:
 - Try compensation settings 1 through 3.
 - Increase the DIB capacitance.

Whenever possible, try to accommodate 25% to 50% more DIB capacitance in your DIB design than what you know you need.

- Check your DUT sense wiring on the DIB.

DIB capacitance DC sensing is the most stable, and AC sensing is next. Die sensing may prove unstable to use.

- Check capacitor selection on DIB layout (use ceramic X5R or X7R type, 10F maximum single value, and package 1210, 1206, or 0805).

- Use the minimum capacitance necessary for the desired performance.

Even though the DCVS supports more than 1000F with a single supply, the trade-off is giving up DIB space and increasing current measurement noise with increasing DIB capacitance. This happens with all power supplies, not just the DCVS.

- Compensation does not affect DC accuracy.

For more information, see [VSM DIB Design Guidelines](#).

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VSM DCVS Programming : DCVS VBT Programming : Setting Alarms

Setting Alarms

An alarm indicates a hardware malfunction or a hardware condition that may invalidate a measurement. These conditions can be caused by a bad device, an incorrectly programmed instrument, or a test system failure. By default, a test is forced to fail if an alarm occurs. You can set the action to be taken for each DCVS pin when an alarm occurs. The test can be forced to bin, or the alarm can be ignored. For more information, see Kelvin Sensing and Alarms Subsystem.

To set up alarms for a test program, use the following VBT syntax:

TheHdw.DCVS.Pins(Pinlist).Alarm(AlarmType) = AlarmBehavior

The following table shows the DCVS instrument alarm types for the VSM.

VSM DCVS Alarm Types for VSM

Alarm Type	Description
All	Activates all DCVS alarms.
Open Kelvin DIB	Monitors the voltage drop between the DCVS and the DIB sense points. It has a programmable Kelvin limit and is preset for 250mV of drop before the alarm.
Open Kelvin DUT	Monitors the voltage drop between the DIB sense points and the DUT sense points. It has a programmable Kelvin limit and is preset for 250mV of drop before the alarm.
Open Sense DUT	Monitors the voltage drop between the DIB sense points and the DUT sense points. It is triggered if the DUT sense connection is open.
Sink Fold Current Limit Timeout	Indicates that the sink fold current limit has timed out.

Source Fold Current Limit Level	Indicates that the source fold current limit level has been exceeded.
Source Fold Current Limit Timeout	Indicates that the source fold current limit has timed out.
Source Overload Current Limit Level	Indicates that the source overload current limit level has been exceeded.
Source Overload Current Limit Timeout	Indicates that the source overload current limit has timed out.
Widened Spec	Indicates that the board is outside of its optimal operating temperature. The board can still operate, but it cannot meet its specification.

The following table lists DCVS instrument alarm behaviors. For more information, see [Programming Instrument Alarm Behavior](#).

DCVS Alarm Behaviors

Alarm Behavior	Description
tlAlarmContinue	Report the alarm but do not change DUT status.
tlAlarmDefault	Restore behavior to default.
tlAlarmForceBin	Bin the current part.
tlAlarmForceFail	Current part fails.
tlAlarmOff	Disable alarms if possible.

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[VSM DCVS Programming](#) : [DCVS VBT Programming](#) : Setting the DCVS Current Meter

Setting the DCVS Current Meter

For more information on setting up the DCVS current meter, see [Current Meter Setup, Strobe, and Readback Programming](#).

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[VSM DCVS Programming](#) : DCVS Pattern Control

DCVS Pattern Control

This section describes how to create DCVS test patterns by writing your own test patterns in an ASCII text file using the digital pattern format and DCVS microcodes. You can also use parameter sets (PSets) to program a DCVS instrument's state for testing.

Use the PatternTool to display and debug DCVS test patterns. Bring up PatternTool in either of the following ways:

- [Click](#) Pattern Tool in the One-Click Launch group on the IG-XL Tools tab.
- [From the Windows](#) Start menu, select Programs>Teradyne IG-XL <version>>Tools>Pattern Tool.

This section includes the following topics:

- [VSM DCVS Pattern Microcodes](#)
- [DCVS Pattern File Example](#)
- [PSets](#)



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VSM DCVS Programming : DCVS Pattern Control : VSM DCVS Pattern Microcodes

VSM DCVS Pattern Microcodes

To create simple or complex test patterns, you can write your own ASCII text file using the digital pattern format and DCVS microcodes. DCVS microcodes can be used to control DCVS settings during pattern execution. This section describes pattern microcodes that are used only in DCVS patterns.

In DCVS patterns:

- All microcodes in a pattern vector are executed when the pattern vector is run. If a microcode in a pattern does not have all of its fields programmed, the unprogrammed fields are programmed to the default values when the pattern vector is run.
- You can have more than one DCVS microcode for a pin in the same pattern. However, you cannot have more than one microcode in the same vector.
- Microcodes are not case sensitive.

Use the following DCVS microcodes in test pattern files:

VSM DCVS Pattern Microcodes	
Microcode	Description
Nop	No operation (does nothing).
Pset <PSet Name>	Applies all parameter settings for the specified PSet to the hardware.
Gate_On	Gates the DCVS on.

Gate_Off	Gates the DCVS off.
Strobe	Strobes the DCVS meter once for that pin or channel. The reading is stored in measurement RAM. To retrieve use the call <code>TheHdw.DCVS.Pins("pinname").Meter.Read(tlNoStrobe, 1)</code> . By specifying no strobe, you read the measurement stored in memory without taking a new measurement.
Enable_Inst_Cond	Turns on signaling of selected DCVS condition bits.
Disable_Inst_Cond	Turns off signaling of selected DCVS condition bits.
Enable_Alarm	Enables reporting of the DCVS source fold current limit level, source overload current limit level, and source overload timeout alarms.
Disable_Alarm	Disables reporting of the DCVS source fold current limit level, source overload current limit level, and source overload timeout alarms while the PatGen is running.
VS_Main_Sel	Selects the main voltage. This value is the main voltage value you programmed before the pattern started, unless a PSet has been applied, in which case that voltage value is used.
VS_Alt_Sel	Selects the alternate voltage. This value is the alternate voltage value you programmed before the pattern started, unless a PSet has been applied, in which case that voltage value is used.
Stop	Stops any capture that is currently happening on the pin.
Start_Cap_Timer	Reserved for future use.
Stop_Cap_Timer	Reserved for future use.
Trig <Signal Name>	Triggers a capture with the specified capture tag and the current hardware settings. During pattern execution, the LoadSettings method cannot be called. So, make sure to set up the hardware as desired before the pattern starts. You can set up capture hardware settings during a pattern burst using PSets. If you do not define the signal name, Trig captures to the default signal as specified in the language.
Filter_On	Turns the meter filter on and the bypass off. When the meter filter is turned on, IG-XL uses the last active filter setting. In VBT, this microcode is equivalent to <code>TheHdw.DCVS.Pins(PinList).Meter.Filter.Bypass = False</code>

Filter_Off	Turns the meter filter off and the bypass on. Note that an analog filter is always in effect on VSM, so meter data is never completely unfiltered. In VBT, this microcode is equivalent to TheHdw.DCVS.Pins(PinList).Meter.Filter.Bypass = True
------------	--

For more information, see [Microcode \(Opcodes and Control Bits\)](#).

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VSM DCVS Programming : DCVS Pattern Control : DCVS Pattern File Example

DCVS Pattern File Example

The following DCVS pattern file shows the required pattern file statements. Note the following guidelines.

To make porting patterns between digital instruments easier, do not include a `digital_inst` or `opcode_mode` compiler options in your pattern file. Instead, include these options in the compile scripts or batch files. By following this programming practice, pattern files are not locked into a particular instrument or timing mode and can be ported easily.

You can execute VSM pattern microcodes from both vector memory (VM) and subroutine memory (SRM).

For an example of a pattern file used in Vbump testing, see [Vbump Testing](#).

Settle Time Repeat Guidelines

The number of repeat cycles used in a settle time repeat depends on the following:

- The timeset period defined in the active time sheet
- The microcode or PSet settle time
- The instrument's analog settling time (if applicable)
- The minimum execution time required between microcodes or PSets

The goal is to have enough repeats so that no microcode is executed before the previous one is completed and/or the new instrument's analog parameters have time to settle (for example, output voltage level). The instrument board specification document specifies the microcode and PSet execution times.

```
import tset ts1;

instruments = {
    vdd : DCVS;
    vio : DCVS;
}

srm_vector ($tset, dig1:S)
{
    > ts1 1;
    > ts1 1;
    // __ GATE ON __
    (vdd : DCVS = Gate_On)
    > ts1 1;
    // settle time repeat. See guidelines above.
    repeat 5000
    > ts1 1;
    > ts1 1;
    > ts1 1;
    > ts1 1;
    // __ GATE OFF __
    (vio : DCVS = Gate_Off)
    > ts1 1;
    // settle time repeat. See guidelines above.
    repeat 5000
    > ts1 1;
    > ts1 1;
    > ts1 1;
    > ts1 1;
    // __ TRIGGER CAPTURE: my_capture_signal_label defined in job thru VBT __
    (vdd : DCVS = Trig my_capture_signal_label)
    > ts1 1;
    // settle time repeat. See guidelines above.
    repeat 5000
    > ts1 1;
    > ts1 1;
    > ts1 1;
    > ts1 1;
    // __ MULTIPLE PINS SAME MICROCODE __
```

```
((vdd,vio) : DCVS = Gate_On)
> ts1 1;
// settle time repeat. See guidelines above.
repeat 5000
> ts1 1;
> ts1 1;
> ts1 1;
> ts1 1;
// __ MULTIPLE PINS DIFFERENT MICROCODES __
(vdd : DCVS = VS_Alt_Sel)
(vio : DCVS VS_Main_Sel)
> ts1 1;
// settle time repeat. See guidelines above.
repeat 5000
> ts1 1;
> ts1 1;
> ts1 1;
> ts1 1;
// __ SINGLE STROBE __
(vdd : DCVS = Strobe)
> ts_strobe 1;
> ts1 1;
// settle time repeat. See guidelines above.
repeat 5000
> ts1 1;
> ts1 1;
// __ PARAMETER SET (my_pset_1, my_pset_2 defined in job thru VBT __
> ts1 1;
(vdd : DCVS = PSet my_pset_1)
> ts1 1;
// settle time repeat. See guidelines above.
repeat 5000
> ts1 1;
> ts1 1;
> ts1 1;
> ts1 1;
(vdd : DCVS = PSet my_pset_2)
> ts1 1;
```

```
// settle time repeat. See guidelines above.  
repeat 5000  
  > ts1  1;  
  > ts1  1;  
  > ts1  1;  
  > ts1  1;  
  > ts1  1;  
}
```

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VSM DCVS Programming : DCVS Pattern Control : PSets

PSets

Parameter sets (PSets) allow you to program the state of an instrument for testing, such as the force voltage value and force voltage range. You set up its characteristics using its parameters. You define a PSet in VBT language and invoke it using PSet microcode.

Once defined, PSets can be applied through VBT or through patterns using the DCVS PSet microcode. Assuming you have defined a parameter set named "my_pset", the following example shows how to apply the PSet from VBT:

TheHdw.DCVS.Pins("vdd").Psets("my_pset").Apply

The statement above is equivalent to:

TheHdw.DCVS.Pins("vdd").Psets.Item("my_pset").Apply

PSet parameters are initialized with their default values, and all parameters are updated when the PSet is applied.

The following example shows an ASCII test pattern with a DCVS microcode entry to apply this parameter set:

```
(vdd:DCVS = Pset my_pset)
> tset0  H L 1 0 . . ;
```

Other PSet language properties and methods:

TheHdw.DCVS.Pins.PSets.Count	This property returns the number of PSets that have been defined for the pin.
TheHdw.DCVS.Pins.PSets.List	This property returns a string array that is a list of all PSet names defined for the pin.
TheHdw.DCVS.Pins.PSets.Add	This method creates a new PSet in PSet memory and sets all PSet parameters to the default settings. If your new PSet has

the same name as a PSet that already exists in memory, then this PSet is set to default settings.

TheHdw.DCVS.Pins.PSets.Item

This property accesses a PSet from the PSet memory, but it does not change that PSet's parameters to default settings. If the PSet does not exist in memory when called by Item, IG-XL returns a run-time error. You can program one or all PSet parameters using this property.

For a PSet example using VBT, see the DCVS PSet example function below.

DCVS PSet Example Function

This example function, AddPSet, creates a DCVS parameter set. To use the AddPSet function in Vbump testing, see [Vbump Testing](#).

```
Function AddPSet(PinName As String, _
    PSetName As String, _
    BandwidthSetting As Double, _
    CurrentRange As Double, _
    SampleRate As Double, _
    SampleSize As Long, _
    SourceFoldLimit As Double, _
    SourceOverloadLimit As Double, _
    MeterCurrentRange As Double, _
    MeterVoltageRange As Double, _
    MeterMode As tIDCVSMeterMode, _
    MainVoltage As Double, _
    AltVoltage As Double _
) As Long
```

```
Dim MinILimitLevelDelta As Double
```

```
On Error GoTo ErrorHandler
```

```
Call TheHdw.DCVS.Pins(PinName).PSets.Add(PSetName)
```

```
With TheHdw.DCVS.Pins(PinName).PSets.Item(PSetName)
```

```
.BandwidthSetting.Value = BandwidthSetting
```

```
' Need to set output current range first because it determines the
```

```
' available Ilimit and Imeter ranges
.CurrentRange.Value = CurrentRange
```

```
.Capture.SampleRate.Value = SampleRate
.Capture.SampleSize.Value = SampleSize
```

```
With .CurrentLimit.Source
```

```
    MinILimitLevelDelta = GetMergeFactor(PinName) * VSM_MIN_ILIMIT_LEVEL_DELTA
```

```
' Note: GetMergeFactor() and VSM_MIN_ILIMIT_LEVEL_DELTA are defined elsewhere
```

```
If SourceOverloadLimit >= SourceFoldLimit + MinILimitLevelDelta Then
```

```
    If SourceOverloadLimit > .FoldLimit.Level.Value + MinILimitLevelDelta Then
```

```
        .OverloadLimit.Level.Value = SourceOverloadLimit
```

```
        .FoldLimit.Level.Value = SourceFoldLimit
```

```
    Else
```

```
        .FoldLimit.Level.Value = SourceFoldLimit
```

```
        .OverloadLimit.Level.Value = SourceOverloadLimit
```

```
    End If
```

```
Else
```

```
' This is an invalid setup
```

```
TheExec.DataLog.WriteComment ("AddPSet() Warning: Cannot set " & _
```

```
    "Source OverloadLimitLevel < FoldLimitLevel + " & _
```

```
    MinILimitLevelDelta & _
```

```
    " for pset named, " & _
```

```
    PSetName & _
```

```
    " (" & PinName & ")." & _
```

```
    " Attempt aborted.")
```

```
End If
```

```
End With
```

```
.Meter.CurrentRange.Value = MeterCurrentRange
```

```
.Meter.VoltageRange.Value = MeterVoltageRange
```

```
.Meter.Mode = MeterMode
```

```
.Voltage.Main.Value = MainVoltage
```

```
.Voltage.Alt.Value = AltVoltage
```

```
End With
```

```
Exit Sub
```

```
ErrorHandler:
```

```
    Dim Reply As Integer
```

```
Reply = MsgBox("Error detected during AddPSet(). Stop to Debug?", vbYesNo, "VSM PSet  
Checkout")
```

```
If Reply = vbYes Then
```

```
    On Error GoTo ErrorHandler
```

```
    Stop
```

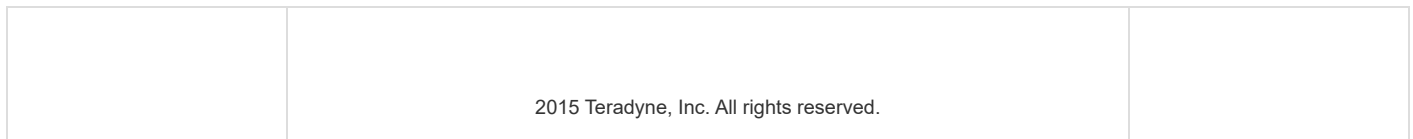
```
Else
```

```
    On Error GoTo 0
```

```
End If
```

```
Resume Next
```

```
End Function
```



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VSM DCVS Programming : DCVS Test Instance Programming

DCVS Test Instance Programming

You can use the instance editor to create and edit DCVSPowerSupply_T VBT test instances. A test instance defines a test that is applied to the DUT when the test program runs. You can create as many test instances as you want. The Test Instances sheet lists all test instances for your test program.

Before creating a DCVS test instance, you must declare a Power pin type in the pin map. You also need a Pin Levels sheet. Finally, you must connect the Power pin to a DCVS channel type.

To create or edit a DCVSPowerSupply_T test instance in the instance editor, see [Creating and Editing Test Instances](#). Once you have created a test instance, program the DCVSPowerSupply_T test instance by filling in the parameter fields. For more information on DCVSPowerSupply_T test parameters, see [DCVS Power Supply VBT Test](#).



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[VSM DCVS Programming](#) : Using VSM in Concurrent Testing

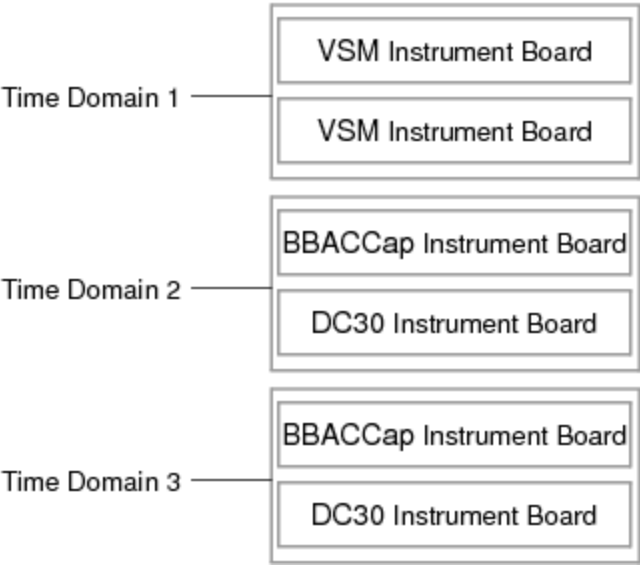
Using VSM in Concurrent Testing

To support concurrent testing, you can assign VSM channels to a time domain. This means that VSM instruments can be part of a concurrent pattern burst. However, only one pattern in the concurrent burst can contain VSM instruments in it at one time. This limitation means that VSM channels cannot be in more than one time domain for each concurrent burst.

Concurrent features for VSM include:

- A VSM board can be configured for one time domain
- Multiple VSM instruments can be grouped into a common time domain
- VSM is restricted to using the Global flow domain and synchronous pattern starts

The following figure shows a configuration with two VSM instruments assigned to its one time domain and other UltraFLEX instruments assigned to other time domains.



VSM Concurrent Testing Configuration Example
For more information, see [About Concurrent Test](#).

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VSM Theory of Operation

VSM Theory of Operation

This section provides a theory of operation that includes a VSM hardware summary, test head connection information, a functional description with block diagrams, and output power specifications.

The following topics are available:

- [VSM Board Hardware](#)
- [Test Head Connections](#)
- [Functional Description: VSM Hardware Design](#)
- [Functional Description: Programmer’s Guide to DCVS Hardware](#)
- [Output Power](#)

For information on VSM instrument performance specifications and standards, see [VSM Specifications](#).

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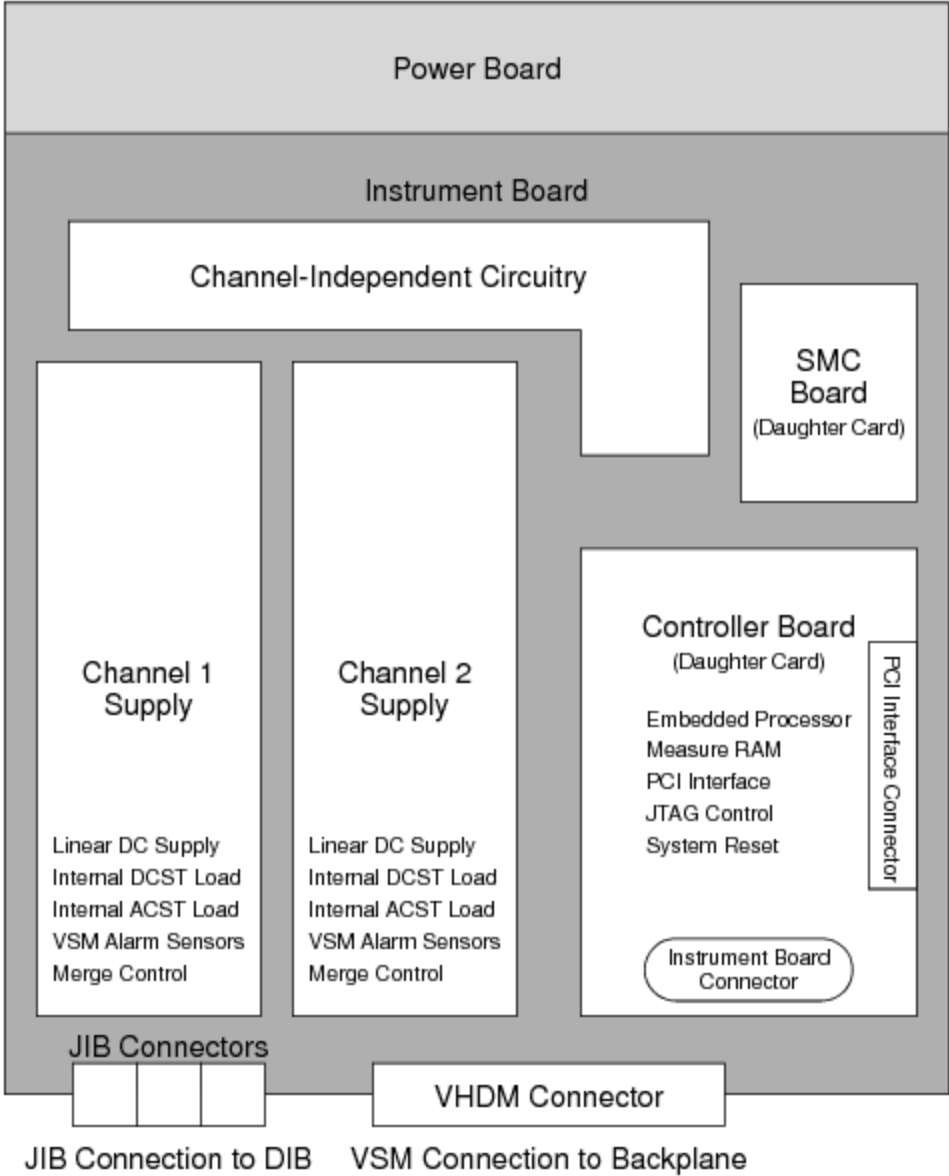
VSM Theory of Operation : VSM Board Hardware

VSM Board Hardware

The VSM instrument board assembly is located in the test head and consists of a cold plate on one side and a set of four boards:

- Controller board
- Power board
- Instrument board
- System Monitor Control (SMC) board

The following figure shows the floorplan of these components on this instrument board.



VSM Instrument Board Hardware Block Diagram
Controller Board

The Controller Board provides the physical means for controlling the VSM. The VSM has built-in software that allows it to carry out user commands and to perform its own diagnostics, calibration, and verification.

The controller board consists of the following components:

- | | |
|---|---------------------------------|
| • | Motorola MPC8245 Microprocessor |
| • | 32 Megabytes of flash ROM |
| • | 128 Megabytes of SDRAM |

- Two Megabytes of Dual-Port SRAM

- VSM Controller FPGA

- Clock and reset circuitry

- External PCI connector

- Instrument Board connector

- JTAG interface

From a software viewpoint, the controller board executes the embedded software. Using IG-XL driver functions, you make an “application call” to the Host Object, which translates the call and passes it to the embedded software. The embedded software then executes the command sequence for the given application call. A command sequence usually involves writes and/or reads to one or more FPGA registers that control some function on the VSM, along with any memory accesses necessary for temporary storage or manipulation of data. Physically, application calls from IG-XL reach the embedded software over the PCI interface bus, which is connected to the microprocessor.

Embedded control of the VSM is done over the processor PortX bus, which is connected to the VSM Controller FPGA. The VSM Controller FPGA translates the PortX addresses to VSM register addresses and writes or reads data to or from those registers. The VSM is controlled by these direct writes and reads. The return value of any host object call is a status indicating the success or failure of the call.

The JTAG interface serves as:

- In-circuit emulator interface for the microprocessor
- Programming interface for configuring the FPGAs of the VSM
- Board-level boundary-scan test factors

Power Board

The power board converts the +48V of system power to all VSM internal power needs. Power is sent to the controller and instrument boards for their own use. The main outputs for channels A

and B are also developed on the power board. A switching power supply is used to produce the main power for these channels. The power board provides preregulated DC-DC supply rails to reduce the overall power dissipation necessary to support a high current linear supply.

Instrument Board

The instrument board consists of two linear (0 to 4V) transconductance power supplies with source and sink current capability. Each supply changes from a current source and sink to a voltage source at the supply sense point. This sense point must be at a low impedance capacitor bank for stable operation. Each supply is independent and supports up to 81A of current. The supplies can be used in a merged configuration (placing them in parallel) with other VSM supplies to increase the total output current by “n” number of merged supplies. In a merged group, one supply becomes the master controller, and the other supplies’ output sections become part of the merged group.

☑ **Note:**

Supplies cannot be stacked in a series merge to increase the output voltage.

The VSM instrument board consists of board level functions as well as channel functions. The instrument board contains the static current and resistive loads (the DCST load) and the dynamic, high-speed loads (the ACST load). These loads provide a high degree of accuracy for calibration. As a result, you can calibrate the VSM internally without requiring a special DIB.

System Monitor Controller (SMC) Board

All modules occupying a slot in the test head require the monitoring and control of the System Monitor Controller (SMC). The SMC protects the test system from hazardous conditions by controlling power sequencing, monitoring faults, and reporting them. If a fault occurs that is hazardous to an instrument board, the overall test system, a DUT, or a user, the SMC immediately shuts down the test head and may even turn off power to the entire test system.

The VSM board has an SMC rider board mounted to the instrument section of the assembly that handles detection and reporting of catastrophic faults. Power to the rider board is independent of the power to the instrument board on which it resides. Slot identification bits from the backplane are connected to this SMC rider board, which allows it to identify the location of the instrument board in the test head.

Because the VSM is a high-power instrument that can distribute potentially hazardous current to a DUT or user, the VSM does not wait for SMC communications before shutting down. The SMC also protects the VSM instrument through an internal alarm monitor. If it detects a catastrophic fault, the VSM executes a fast, but orderly shutdown sequence. Specifically, the VSM gates off the voltage to the DUT, disconnects itself from the DIB, and disables its current amplifiers. To assist with tracking of critical VSM system faults, the VSM still reports the fault to the SMC, which in turn, logs the faults for maintenance review.

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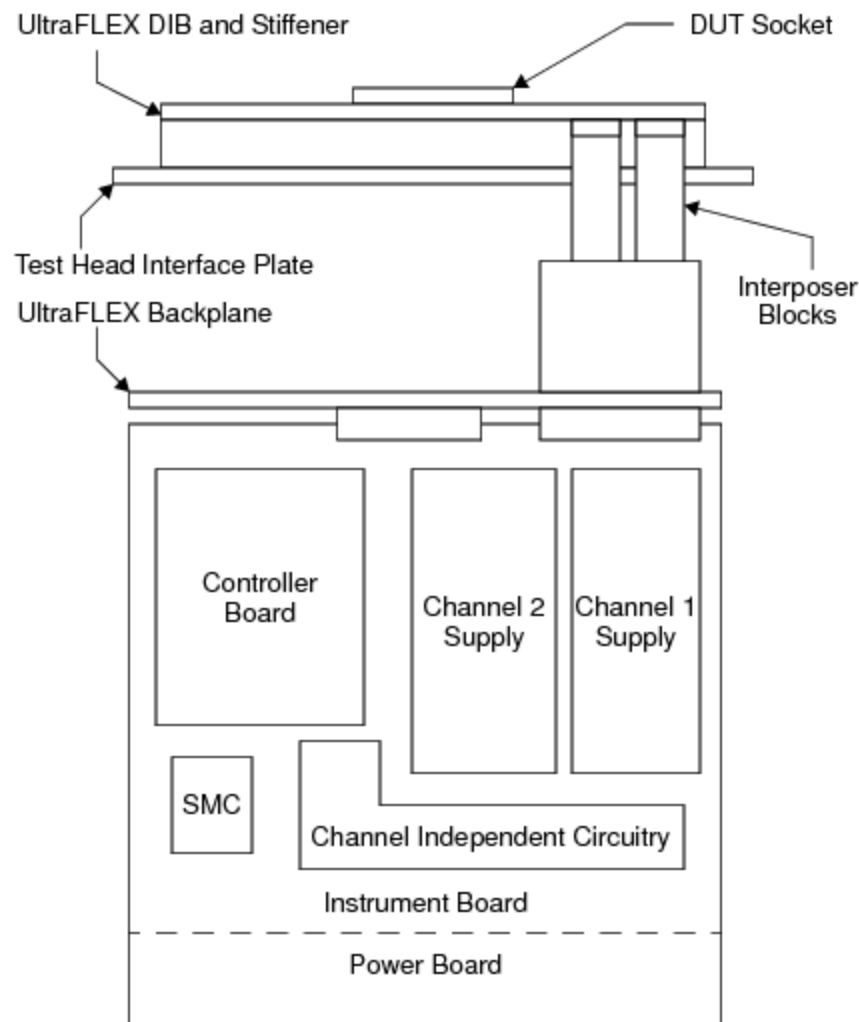
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VSM Theory of Operation : Test Head Connections

Test Head Connections

This section describes where VSM boards are connected to the test head backplane. The following figure shows how the instrument board is connected with an interposer block to deliver power to the DUT.



VSM Instrument Board Connection to the DUT

Several VSM boards can be installed in the UltraFLEX test head. A VSM can be installed in any available slot in the test head. To ensure compatibility across a variety of UltraFLEX test system applications, however, Teradyne recommends the test head and DIB slot locations for up to six VSM boards, as listed in the following table. VSMs are connected to the DIB using three JIB assemblies occupying sidewalks two, three, and four.

UltraFLEX VSM Test Head Configuration

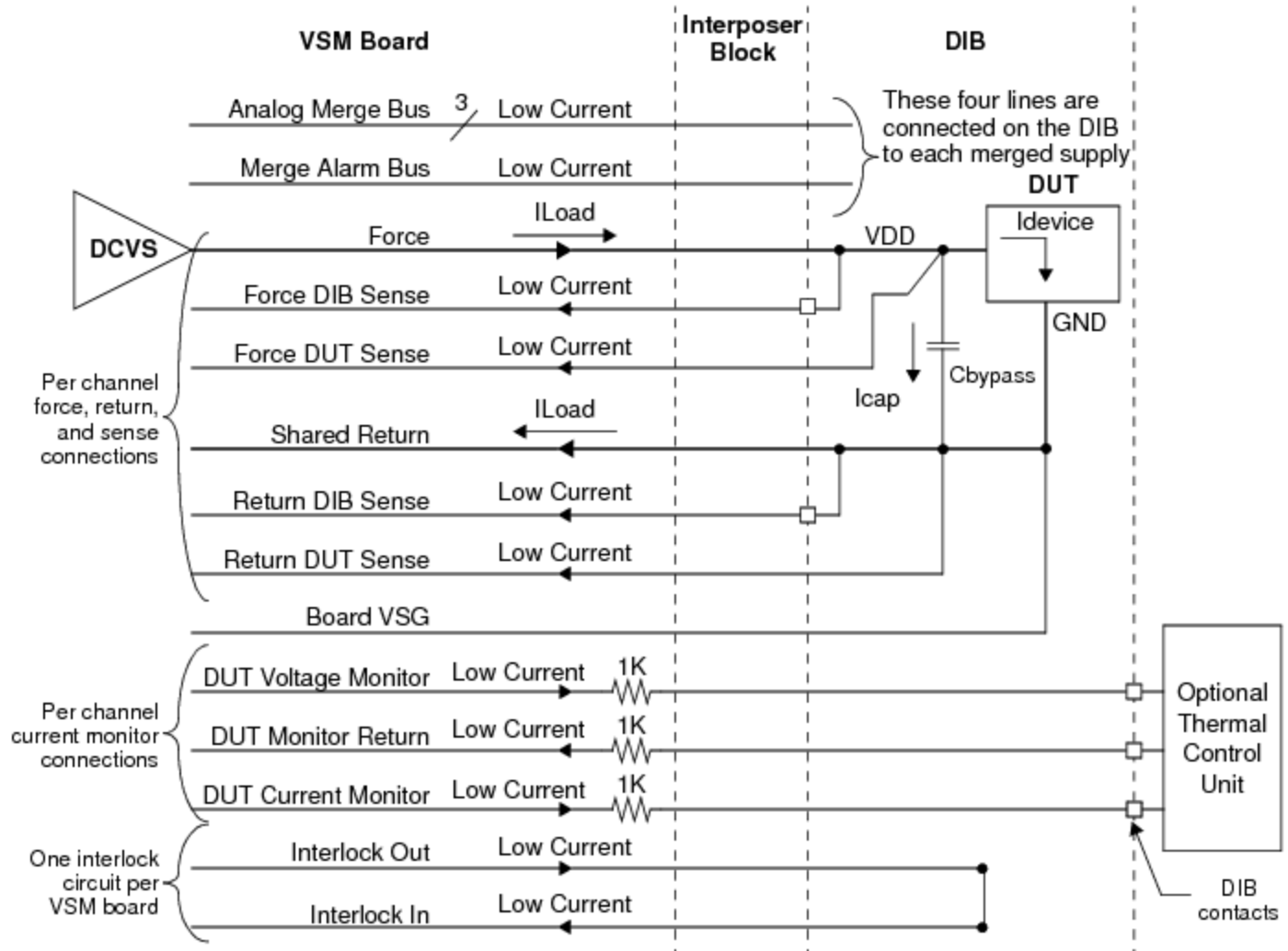
Board Location	Slot	Slot	Board Location
Support VSM 4	37	34	VSM 5
	33	30	
	29	26	
	25	22	VSM 1
	21	18	
	17	14	
	13	10	
	9	6	
	5	2	
	1	0	
VSM 3	3	4	
	7	8	
	11	12	
	15	16	
	19	20	VSM 2
	23	24	Support
VSM 6	27	28	
	31	32	
	35	36	

 Note:

For applications requiring more than three VSM instrument boards, contact Teradyne, whose test applications team can help you develop a solution that meets your needs.

The VSM test head interface features a high-performance connection to support high current to the DUT through a low impedance path. This interface supports the ability of each DCVS to respond rapidly to current transients, which helps reduce the need for large amounts of power supply bypass capacitance on the DIB. This saves space on the DIB for additional test sites, further enabling the testing of multiple devices at once. For more information, see the VSM DIB Design Guidelines.

The following figure shows a more detailed connection model for DCVS signals to the DUT.



VSM DCVS Channel Connections to the DUT

✓ **Note:**

For proper operation, follow the VSM DIB design guidelines for DIB construction, connections, capacitance, and sense lines. Failure to do this may cause instability or oscillation at the output.

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VSM Theory of Operation : Functional Description: VSM Hardware Design

Functional Description: VSM Hardware Design

This section describes the hardware that underlies operation of the VSM.

The VSM is universal in placement in the test head, although the preferred locations are shown in [UltraFLEX VSM Test Head Configuration](#). An internal power board converts the +48V input to lower voltages necessary for operation, while internal loads provide for a “no undocking” calibration.

The VSM has two types of connections to the test head:

- | | |
|--|--|
| | <ul style="list-style-type: none">• Three high-current, low-inductance, semirigid cables which contain force and return connections for power and interface signals to the JIB |
| | <ul style="list-style-type: none">• Backplane connections to the tester through a VHDM connector |

Thermal management of the VSM is handled by liquid cooling through a cold plate attached to the central cooling connections of the tester.

A single 16-bit DAC sets the output force voltage from 0V to 4V. VSM supports a Vbump operation and setting of various parameters using parameter sets (PSets). A gate off signal forces the supply to 0V.

Each supply functions as a transconductance amplifier providing the current required by the DUT. The supply’s DC source current has five selectable ranges: 1.0A, 11A, 21A, 51A, and 81A. These supplies can be merged in parallel to provide multiples of these current modes. Up to six supplies on three boards can be merged for a total of 486A. You can tailor the supply’s response for the DIB design, bypass capacitance, and desired performance using five internal compensation settings.

The VSM has two 12-bit source current clamp DACs. They support a two-level source current clamp (constant current) function. The DACs’ voltage level is compared with a voltage level in the supply

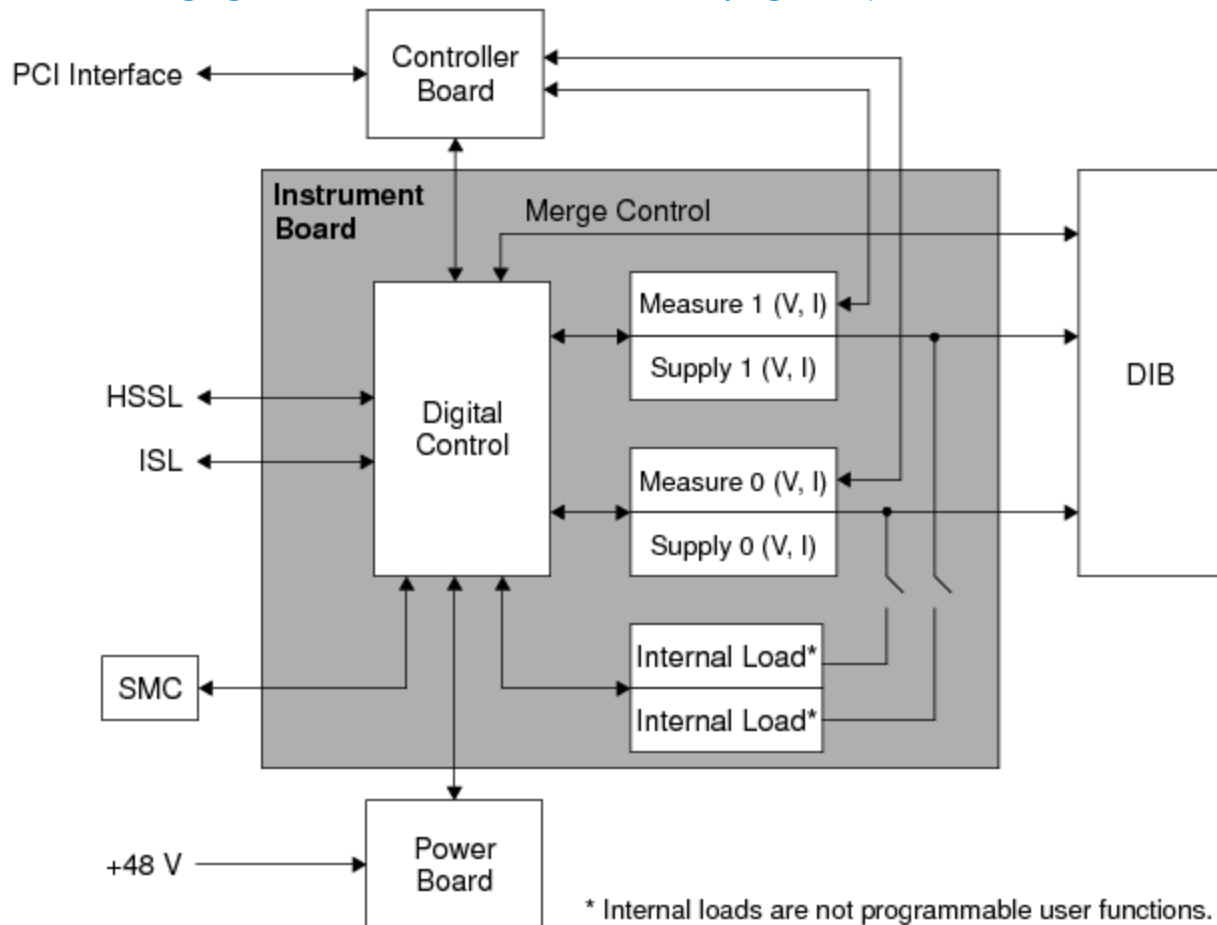
control circuitry that represents the current output. Clamp circuitry holds the output current at this programmed level.

Each level has a programmed time. After the timeout for the first level clamp, the circuitry switches to the second level. After this timeout you can continue forever (constant current mode) or FOLD (Gate OFF or force 0V). This two-level clamp supports higher dynamic currents during parts of the test while switching to a lower level clamp during less dynamic conditions. The clamp levels can be changed using PSets.

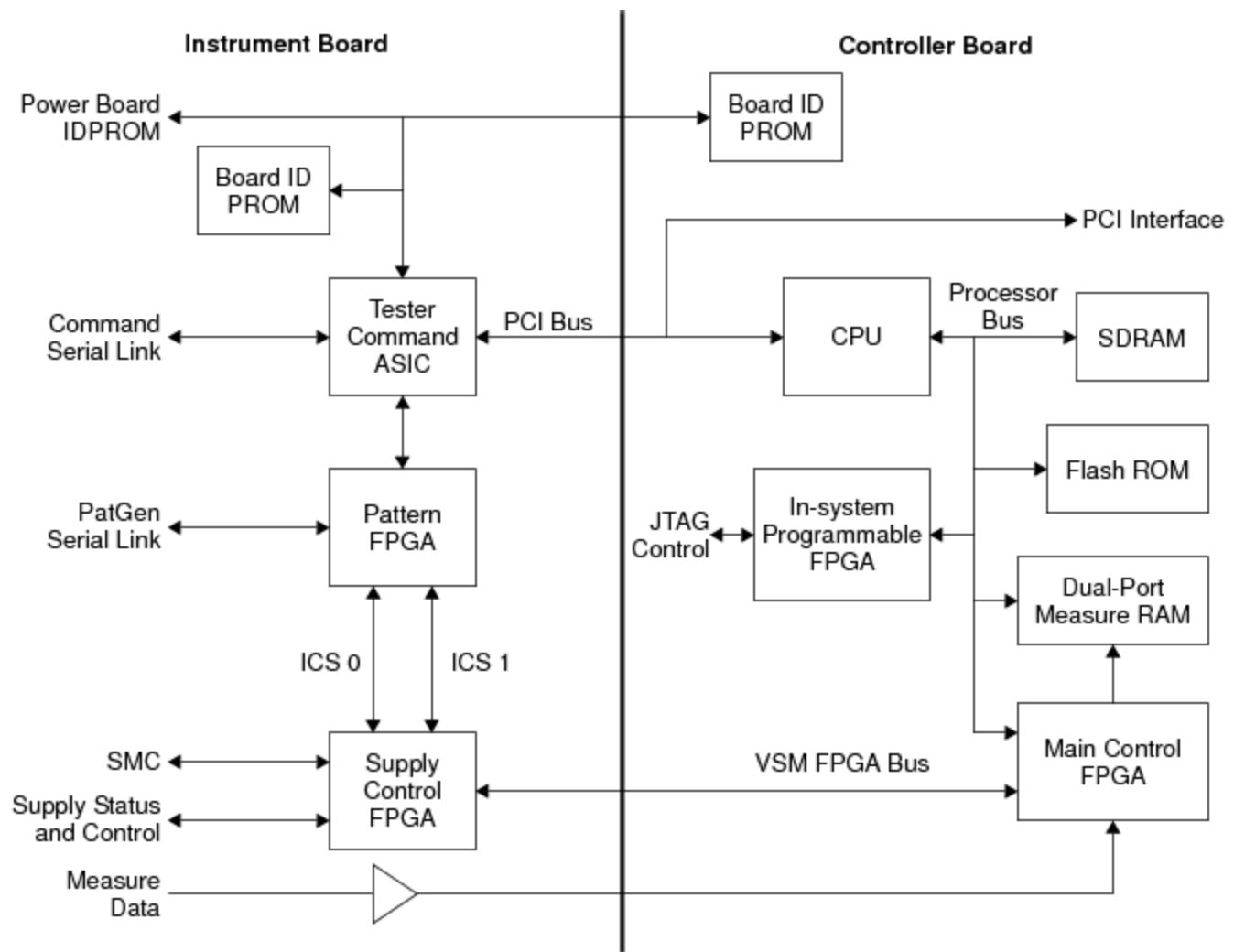
The VSM has two current measurement ranges for each output current range. The instrument allows you to take measurements individually or store them in a Measurement RAM for later processing. Each DC source current range is also a measurement range.

You can connect each supply to the DIB or to an internal load used for calibration. While connected to the DIB, alarms monitor the voltage drop from the supply to the DIB and from the DIB to the DUT. The limit for this DIB-to-DUT voltage drop can be programmed with a 12-bit DAC. The VSM uses a single 24-bit ADC and internal loads during the calibration and validation process to ensure accurate results. The internal loads and the 24-bit ADC are not programmable user functions.

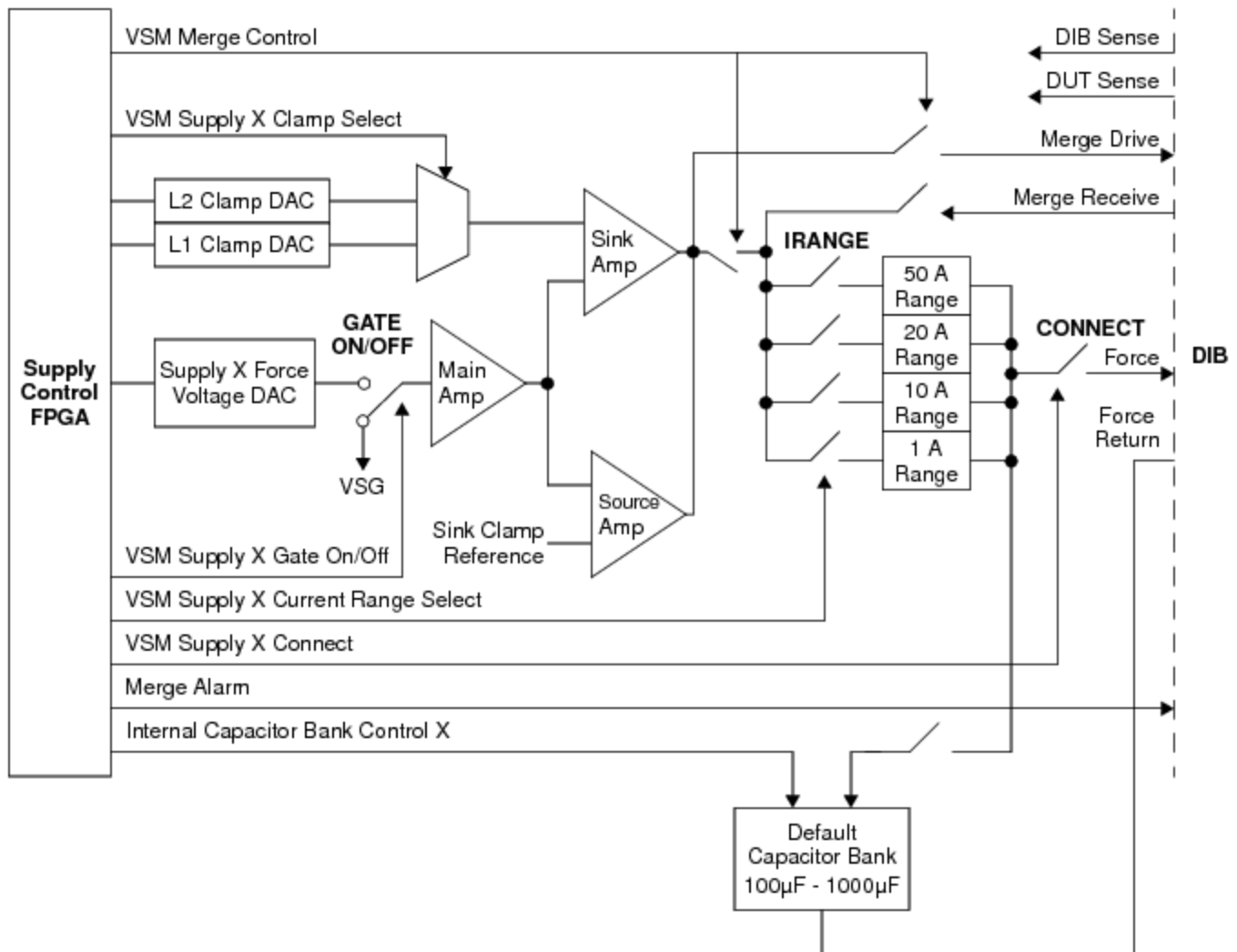
The following figures describe the hardware underlying the operation of the VSM.



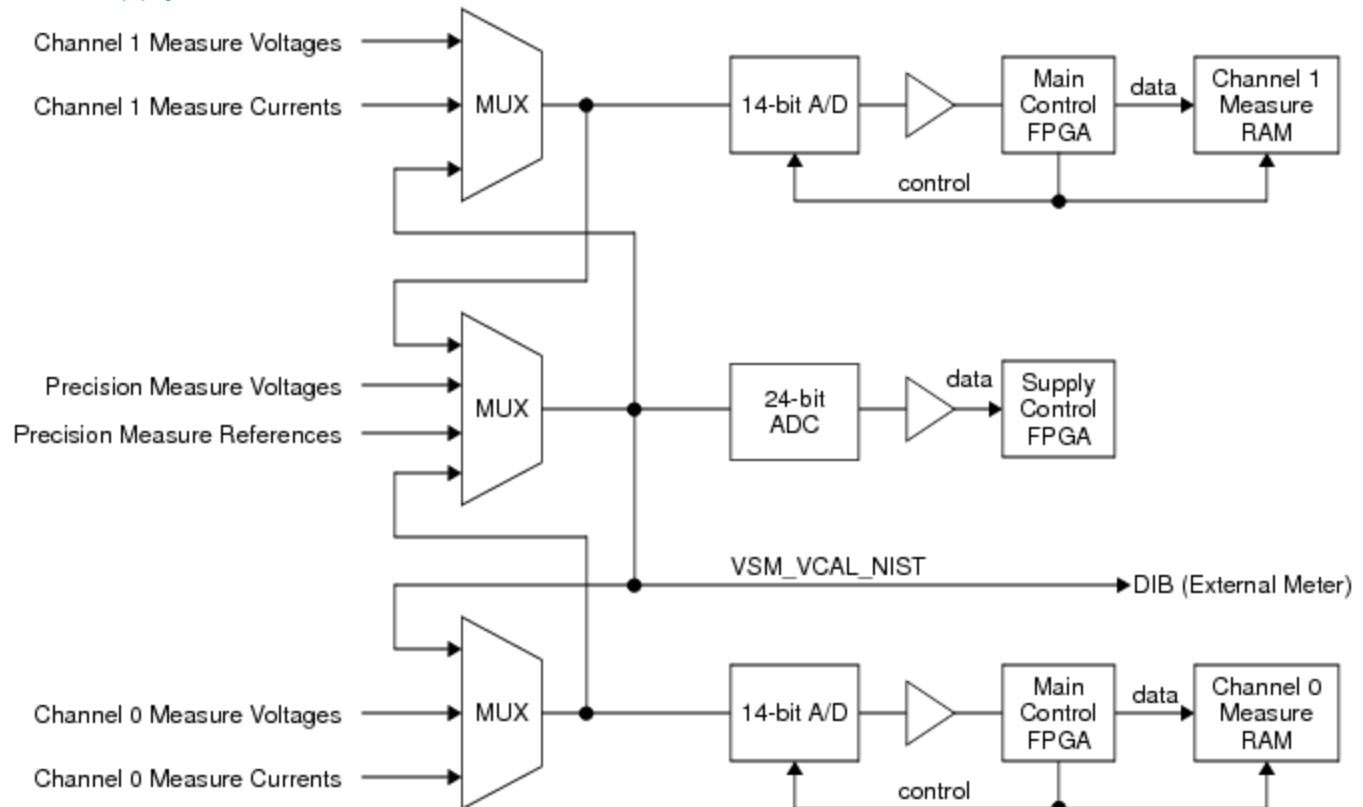
VSM Basic Functional Block Diagram



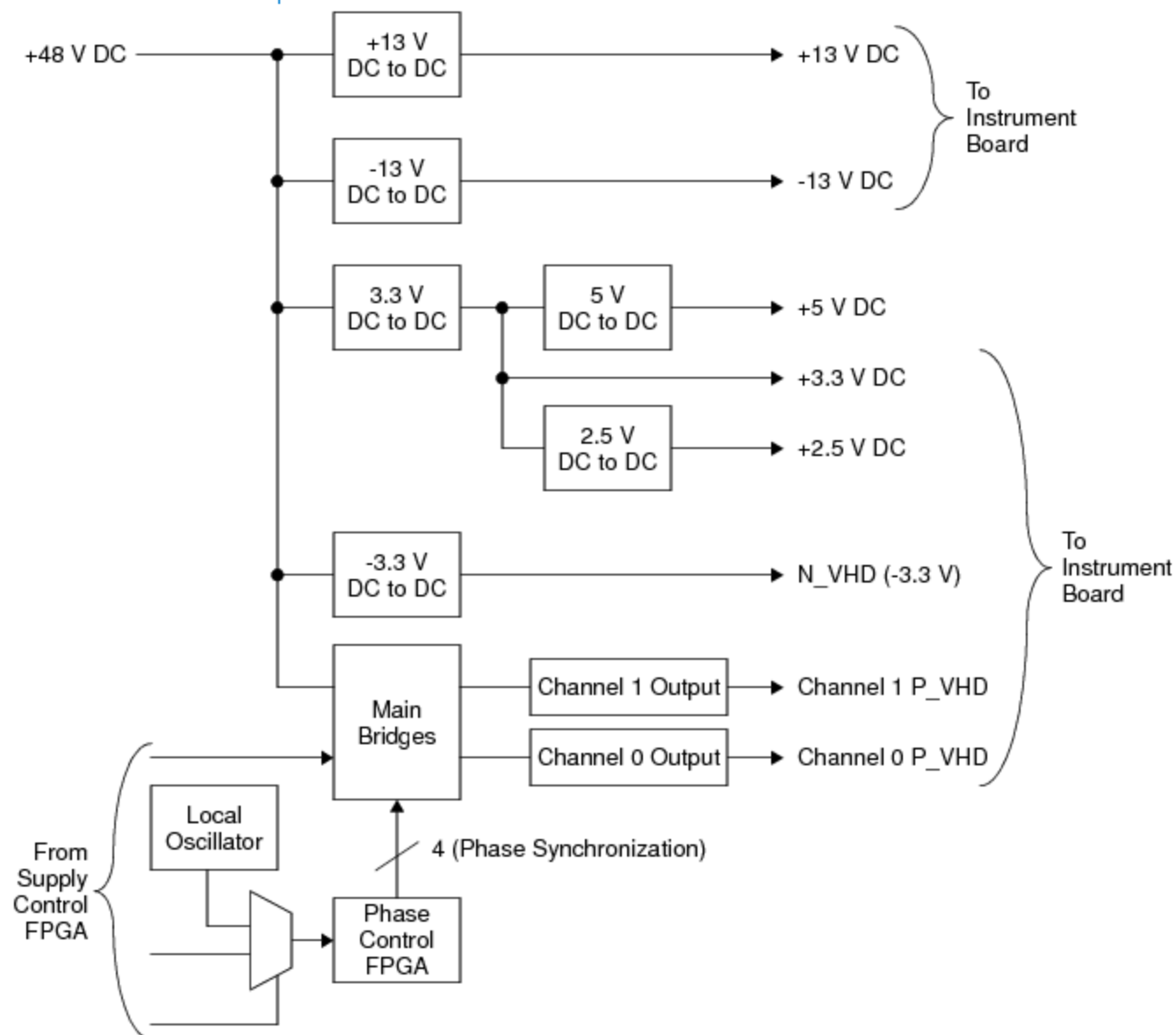
VSM Digital Control Hardware



VSM Supply Hardware (One Channel)



VSM Measure and Capture Control Hardware



VSM Power Board

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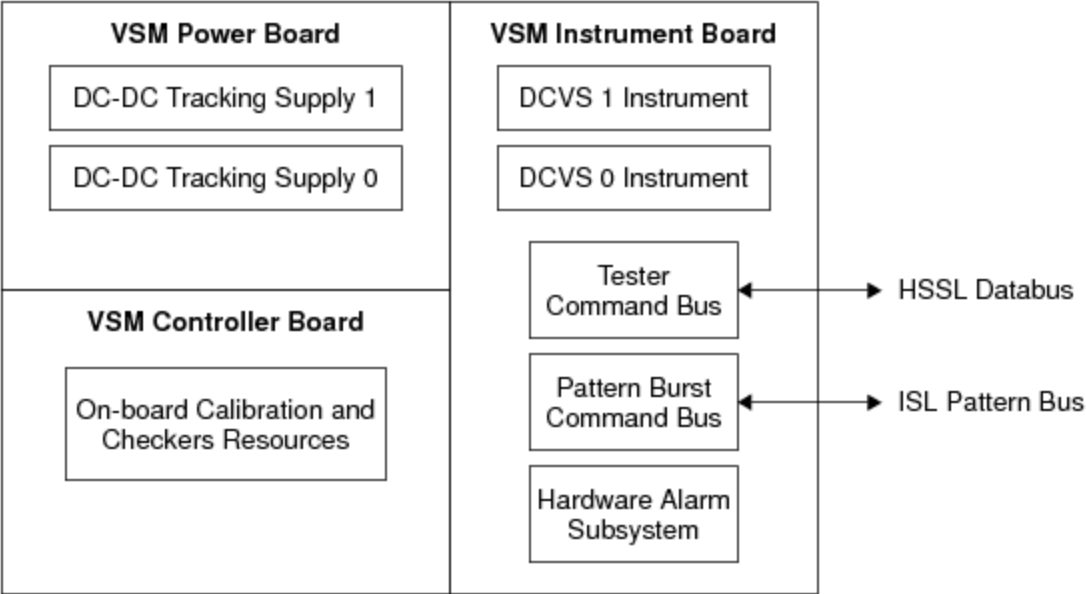
VSM Theory of Operation : Functional Description: Programmer’s Guide to DCVS Hardware

Functional Description: Programmer’s Guide to DCVS Hardware

This section describes how signals move through the VSM instrument board to the DUT from a programmer’s point of view. From a programming perspective, the VSM board can be viewed as a set of one or two independent DCVS instruments. However, the implementation of most programmable DCVS features are distributed across the VSM analog and digital boards. The power board also plays an important role in VSM operation.

The VSM board consists of several functional subsystems, as shown in the following figure:

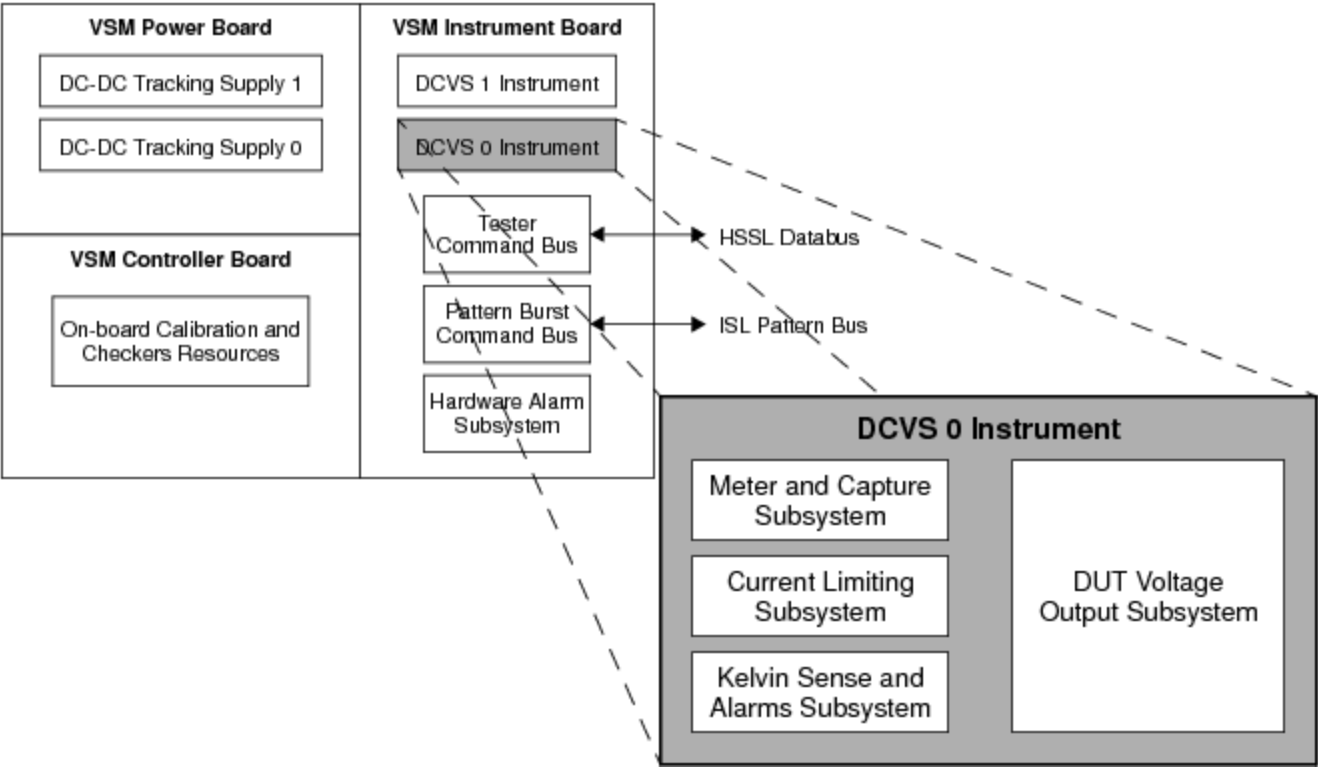
- | | |
|--|---|
| | <ul style="list-style-type: none">Two DCVS instrument subsystems that can be merged to create fewer, more powerful DCVS instruments |
| | <ul style="list-style-type: none">Tester command bus subsystem (high-speed serial link (HSSL) databus) |
| | <ul style="list-style-type: none">Pattern burst command bus subsystem (instrument synchronization link (ISL) pattern bus) |
| | <ul style="list-style-type: none">Hardware alarm detection and response subsystem |
| | <ul style="list-style-type: none">Calibration and checkers subsystem |



VSM Board Functional Overview

Each DCVS instrument consists the following functional subsystems (see figure below):

- DUT Voltage Output Subsystem
- Current Limiting Hardware Subsystem
- Metering and Capture Subsystem
- Kelvin Sensing and Alarms Subsystem



DCVS Instrument: Logical/Functional Overview

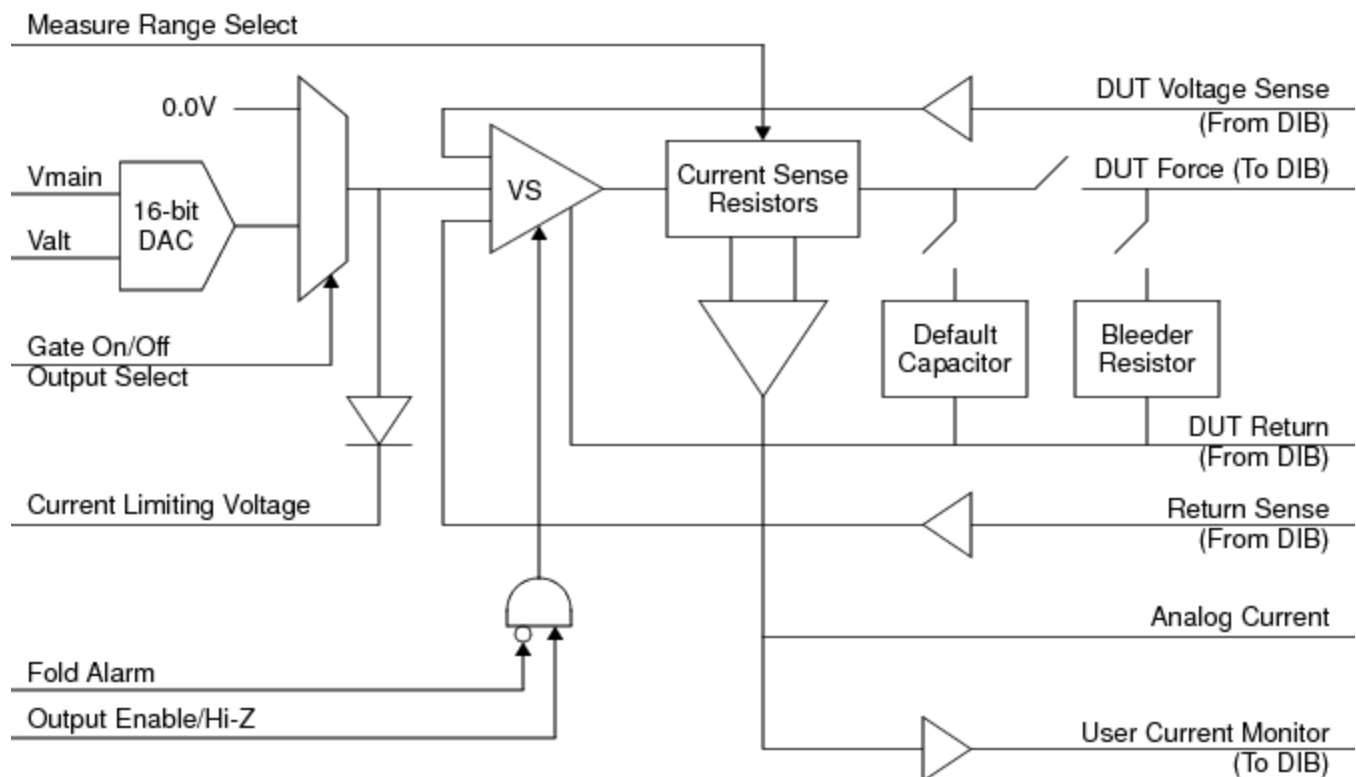
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VSM Theory of Operation : Functional Description: Programmer's Guide to DCVS Hardware : DUT Voltage Output Subsystem

DUT Voltage Output Subsystem

The following figure is a block diagram of the DCVS output subsystem.



DCVS Instrument: DUT Voltage Output Subsystem

Output Voltage DAC

The programmed output voltage is set using the 16-bit DAC, which you can program as a main or alternate voltage. Whether you use the main or alternate voltage, an on/off gate and an output selection multiplexer determines the value of the DCVS output stage. The selected DAC value controls the output driver delivering power to the DIB through the DUT force lines.

DUT Voltage Sense Signal

The DUT voltage sense signal monitors the voltage at the DUT. When the voltage at the DUT exceeds the selected programmed value, the output driver drops its voltage, thereby decreasing the current sourced to the DUT. Similarly, when the sensed voltage at the DUT drops below the programmed value, the output driver increases its output voltage and sources more current to compensate for the drop at the DUT pins. This is an automatic and continuous process that uses the DUT sense feedback to control the voltage delivered to the DUT.

When the DCVS output current changes rapidly from a low level to a high one, you expect a voltage drop. The DCVS senses this drop and compensates for it. The DCVS output system bandwidth controls how quickly the DCVS output responds to changes in the DUT capacitor voltage.

The DCVS also includes programmable bandwidth settings. By changing the bandwidth setting, you can program the DCVS output to minimize droop or lower the voltage and current noise.

Current Limiting Signal

The current limiting signal and fold alarm help to control the maximum current that can be supplied to the DUT. For more information, see [Current Limiting Hardware Subsystem](#).

The DUT force and return lines make up the complete current loop bringing power to the device. They support the full current capability of the force lines. The return sense lines sense the power supply return path voltage at the DUT. So they are also part of the DUT voltage analog feedback control that monitors the voltage across the DUT power pins.

Note:

Return paths are shared between DCVS pairs. This can affect DIB design if you must separate the return planes.

Gating off the DCVS output sets the voltage output value to the voltage value sensed by the VSG DIB input signal (not shown). Ideally, this is the DUT ground voltage of 0.0V.

The analog current and voltage meter circuitry is embedded with the voltage output circuitry. This circuitry includes the buffered DUT voltage sense input, current measure range select signal, analog current measurement hardware, and internal current signal shown in the figure above. For information on the functionality of these components, see [Metering and Capture Subsystem](#).

Voltage and Current Monitor for Thermal Control

The voltage and current monitor are low-current (1K impedance) output signals that are routed to the DIB interposer. With a properly designed DIB, the user voltage and current monitor signals can provide real time DUT power consumption information to an active DUT temperature control unit connected to the DUT side of the DIB.

The voltage monitor tracks the force voltage at the DUT, while the current monitor output is not affected by changes in the current measure range. The current monitor is a voltage representation of the output current. The relationship can be described as:

$$2.37V = 81A * N$$

or:

$$N \times 34.18A / V$$

where N = the number of merged DCVS outputs. Thus, a voltage of 2.37V indicates that the maximum DCVS output current is being delivered to the DUT.

Local Output Capacitor, Bleeder Resistor, and Rapid Discharge

The output subsystem of each DCVS includes a default capacitor and bleeder resistor. When the DCVS is disconnected from the DIB, the default capacitor keeps the DCVS voltage output from oscillating, and the bleeder resistor helps quickly discharge DIB capacitance. You can program and control the connection state of this capacitor and resistor. The default states for these components is determined by hardware based on the state of the DCVS connection state.

To support the bleeder resistor, which holds the voltage across the DIB capacitors at 0V, the VSM employs a rapid discharge feature. When the DCVS is gated off, this feature applies a full 80A current sink for approximately 2ms to fully discharge the largest specified DIB capacitance. When discharging large DIB capacitance, this discharge feature can save hundreds of milliseconds of test time.

The default states are outlined in the table below.

Default Capacitor, Bleeder Resistor, and Rapid Discharge Connection States or Actions

Component	Default Connection State (Action)	
	DCVS Output Connected	DCVS Output Disconnected
Default Capacitor	Disconnected	Connected
Bleeder Resistor	Disconnected	Connected
Rapid Discharge	Disconnected (No action)	Connected (Sink 80A for ~2ms)

 Note:

If you manually set the local capacitor to Off when you disconnect from the DUT, the local capacitor always reverts to the Auto mode to ensure the stable operation of the DCVS.

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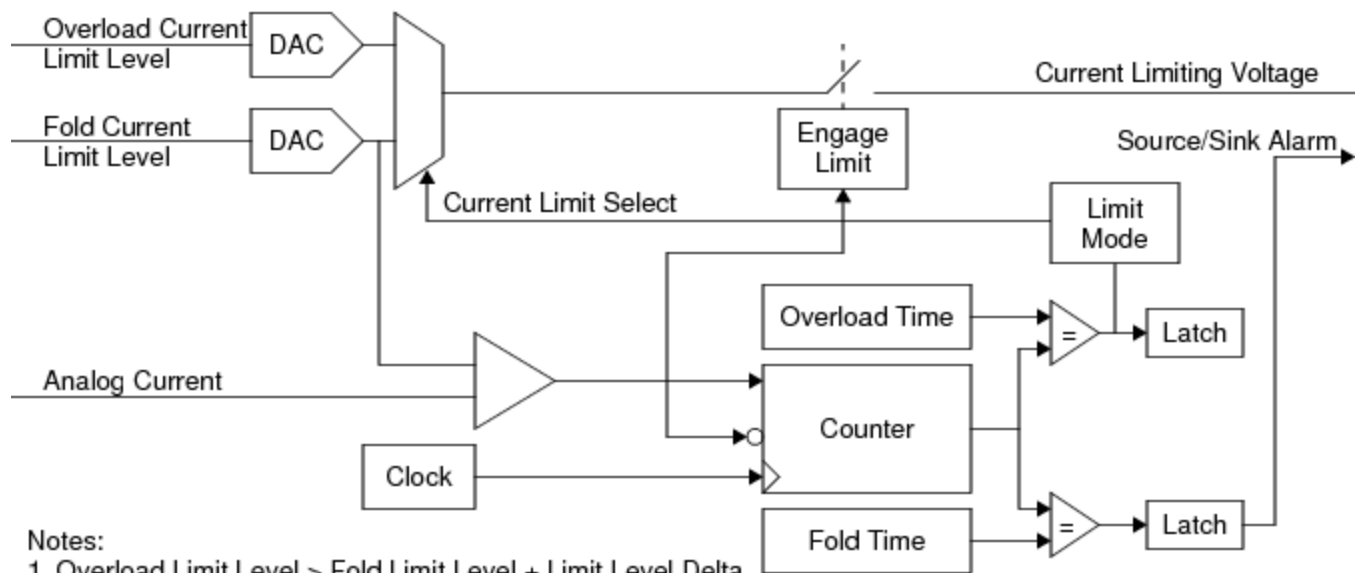
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VSM Theory of Operation : Functional Description: Programmer's Guide to DCVS Hardware : Current Limiting Hardware Subsystem

Current Limiting Hardware Subsystem

Current limit programming consists of setting the values of various DACs and counters. The following figure illustrates the concept behind a single DCVS instrument's source current limiting hardware.



Notes:

1. Overload Limit Level > Fold Limit Level + Limit Level Delta
2. Overload Time < Fold Time
3. Programmed Fold Limit Level is subject to output and adjacent pair power compliance restrictions.
4. Limit Level Delta is specified in the VSM instrument board specification document.

DCVS Instrument: Source Current Limiting Subsystem

The sink current limit is fixed at approximately 7.5% of the output current mode range, or about 6A in the 81A mode.

While the DCVS is gated on, the analog current output signal is compared against the fold limit level. If the analog current exceeds the programmed fold limit level, the current limiting voltage is engaged and a current limit timer is started.

Depending on the current limiting setting, the output current is limited to either the overload limit level (overload enabled) or the fold limit level (overload disabled). With the overload limit level enabled, if the timer value exceeds the programmed overload limit time value, the DCVS voltage output level is limited to the fold limit level. With the overload limit level disabled, the overload limit time is ignored. In both settings, however, if the timer value exceeds the fold limit time, a fold signal is set and the DCVS is gated off or it is left gated on until the maximum fold limit time value is exceeded. See [Setting the Current Limit](#) for programming guidelines.

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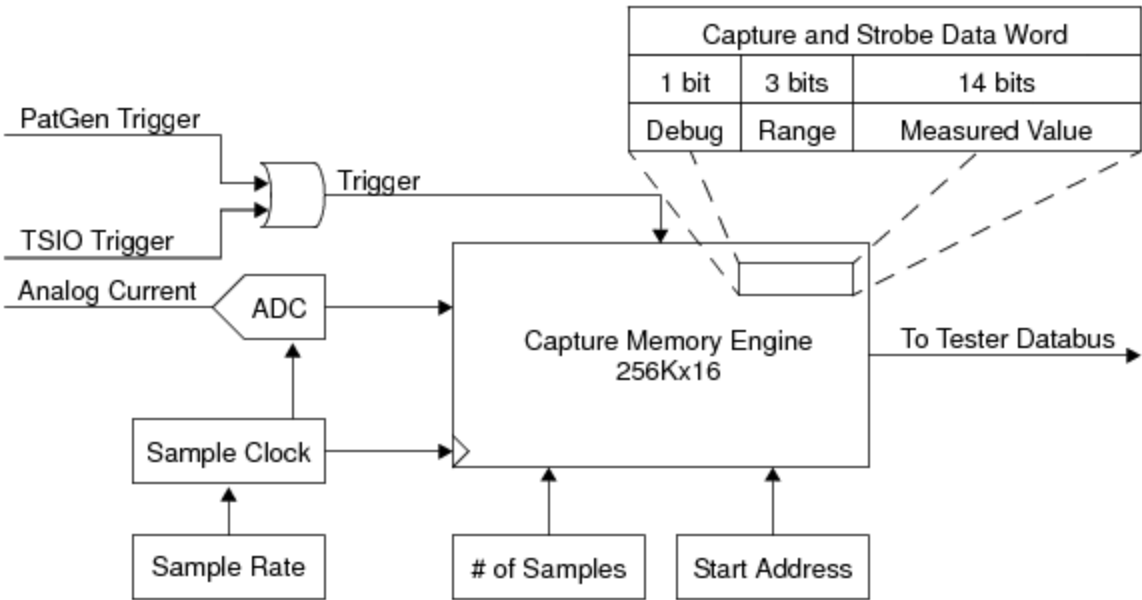
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VSM Theory of Operation : Functional Description: Programmer’s Guide to DCVS Hardware : Metering and Capture Subsystem

Metering and Capture Subsystem

An ADC is at the heart of the DCVS meter. You can use the meter’s ADC to convert either analog current or buffered DUT voltage sense signals to digital values. The digital values can then be stored in the DCVS capture memory. From there, the values can be read back to the host computer through the TSIO databus using the meter and capture programming interfaces. These features are shown in the following figure.



DCVS Instrument: Metering and Capture Memory Subsystem

You must set the meter mode, either voltage or current, to make a meter measurement. The default meter mode is to measure current. Set the desired meter range before taking the measurement. For a single-strobe measurement this is all the setup that is required before reading. To take multiple samples or to capture a waveform, you must program the sample rate and number of samples. When capturing a waveform, you must also program a signal name.

The DCVS supports PSet current mode changes. Note that the current mode range is tied to the current measure range. That is, when you select the current mode range, you also select the current measure range. The current mode range selects the maximum potential DC current available from the VSM. Note that you can set source clamps to limit the DC current further. You must select the proper current mode (therefore, the current measure) range from a PSet during run time. For a list of output current ranges and corresponding current measurement ranges, see [VSM Output Current and Current Measurement Ranges](#).

The ADC clock is fixed and continuous at 31.25MHz. The user-defined sample rate only changes the sample clock that samples the ADC input to the capture memory engine.

DCVS waveform capture can be triggered either through a program command or through a pattern generator microcode.

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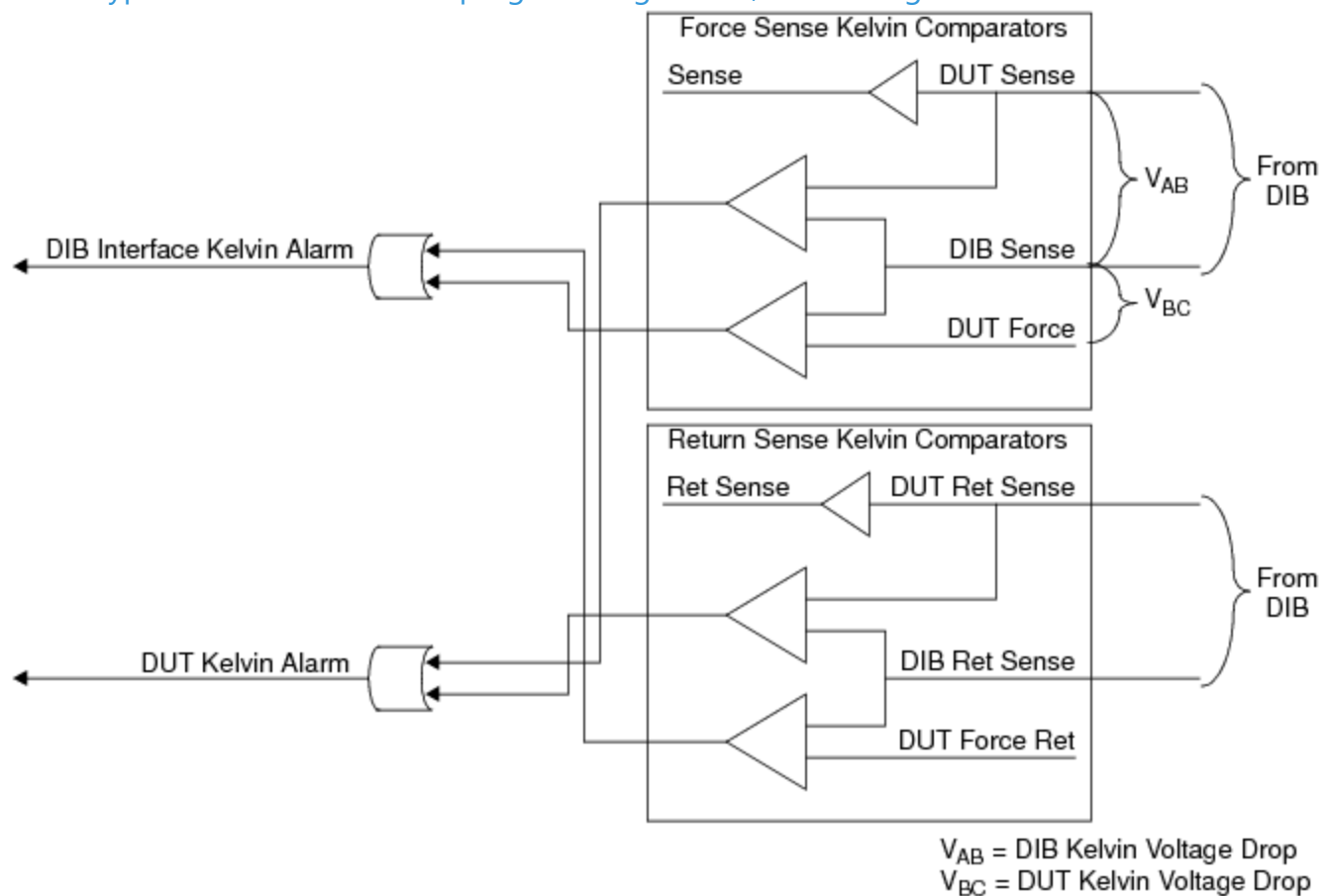
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VSM Theory of Operation : Functional Description: Programmer's Guide to DCVS Hardware : Kelvin Sensing and Alarms Subsystem

Kelvin Sensing and Alarms Subsystem

The following figure shows the DCVS Kelvin sensing and alarm subsystem hardware. For a list of alarm types and information on programming alarms, see [Setting Alarms](#).



DCVS Instrument: Kelvin Sense and Alarms Subsystem

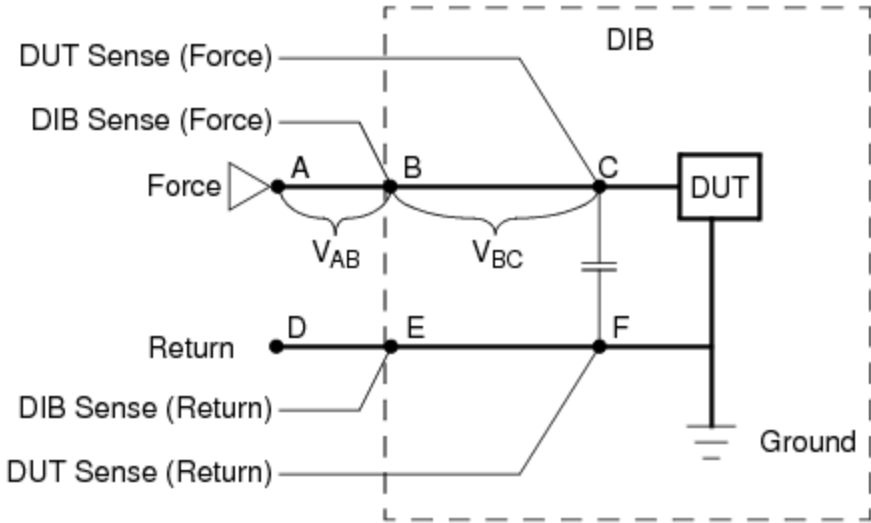
DIB Interface Kelvin Alarm

On a VSM board, each voltage supply requires DIB sense Kelvin connections. These DIB sense pins assure good contact between the DCVS interface (interposer) and DIB signal pads. Two DIB

interface sense pins are used for each DCVS output channel, one for the force path and one for the return path. The DCVS triggers a DIB Kelvin alarm if the voltage difference between the VS output point and the corresponding DIB interface sense point is greater than the preset Kelvin threshold value. (Default is approximately 300mV.) See the figure below.

DUT Kelvin Alarm

Each VSM DCVS channel requires DUT sense Kelvin connections. Two DUT sense pins are used for each DCVS output, one for the force path and one for the return path. The DCVS triggers a DUT Kelvin alarm if the voltage difference between the DIB Kelvin point and corresponding DUT voltage sense point is greater than the preset Kelvin threshold value. (Default is approximately 400mV.) You can adjust this alarm from 0mV to 500mV with a 12-bit DAC. See the figure below.



V_{AB} = DIB Kelvin Voltage Drop
 V_{BC} = DUT Kelvin Voltage Drop
Points B and E are just inside the DIB.
Points C and F are as close as possible to the DUT.

DIB Interface and DUT Kelvin Alarm Sense Points

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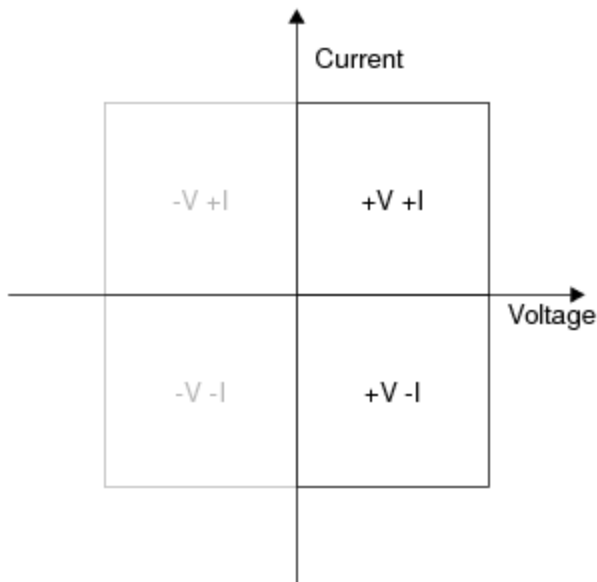
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VSM Theory of Operation : Output Power

Output Power

This section discusses the output power capabilities and restrictions of the VSM board's DCVS instruments.

Each DCVS instrument on the VSM board operates in two quadrants, allowing the sourcing and sinking of current. See the following figure.



Voltage and Current Operating Quadrants

VSM DCVS instruments are not floating voltage supplies. As a result, you cannot stack their outputs to provide greater output voltages. Moreover, you must wire the ground reference (VSG) for each DCVS to the voltage supply return (VS_RTN_n) along with return sense (DUT_RTN_S_n) and at the same voltage as the device ground sense (DGS). For more information, see the [VSM DIB Design Guidelines](#).

Current Sourcing

In single supply mode, the VSM can supply a maximum DC current output of 81A (162W of power) to each of the two channels, 0 and 1, for a combined power of 324W. When merging supplies, the

maximum output power increases as you merge channels:

- | | |
|---|-----------------------------|
| • | 2 VSM channels: 162A (324W) |
| • | 3 VSM channels: 243A (486W) |
| • | 4 VSM channels: 324A (648W) |
| • | 5 VSM channels: 405A (810W) |
| • | 6 VSM channels: 486A (972W) |

The minimum available transient current is greater than 125% of the current mode.
Note that the VSM cannot supply maximum voltage at maximum current, no matter what configuration you are using. When supplying maximum current, which is 81A in single supply mode, the maximum voltage is 2V. This voltage to current relationship is based on the equation $P = VI$. All merge modes are also limited to this power constraint.

Current Sinking

The VSM is designed to source high DC currents. However, it does have a limited capability to sink DC current. A minimum DC current sink of 7.5% of the selected current mode range (approximately 6A in 81A mode) can be maintained continuously. The supply can provide a transient current equal to the source transient current (125% of the current mode) for a period of time to support DIB capacitance charging and discharging.